Task 3 – Flood Hazard Assessment

Regionally consistent risk assessment for earthquakes and floods and selective landslide scenario analysis for strengthening financial resilience and accelerating risk reduction in Central Asia (SFRARR Central Asia disaster risk assessment)

FINAL VERSION

2 September 2022

RED – Risk Engineering + Development – Pavia, Italy

Revision History

Version	Date	State	Initials	Changes
r1d1	05/10/2021	Draft	RED: SD, GB, GC	
r1d2	11/10/2021	Draft	RED: PB; OGS: SP	General review, copyediting
r1	28/10/2021	Approved	RED: PB, GC; OGS: SP	Changes based on reviewers' comments
r2	03/12/2021	Approved	RED: GC, SD	Changes based on reviewers' comments
r3	24/01/2022	Approved	RED: GC, PC, ST	Changes based on reviewers' comments
r4	15/02/2022	Approved	WB	
r5	02/09/2022	Approved	RED: GC, PC	Appendix A – List of Acronyms

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Technical Assignment number 1266456

RED Risk Engineering + Development

Executive Summary

With the goal of improving financial resilience and risk-informed investment planning, the European Union, in collaboration with the World Bank and the GFDRR, has started a program for "*Strengthening Financial Resilience and Accelerating Risk Reduction in Central Asia*" (SFRARR), aiming to advance disaster and climate resilience in Central Asia countries, which includes Kazakhstan, Kyrgyz Republic, Tajikistan, Turkmenistan and Uzbekistan. The program includes several operational components, all contributing to the development of a comprehensive probabilistic risk assessment, consistent across multiple hazards and asset types of the target countries.

Central Asia is an area characterized by complex tectonic and active deformation. The related seismic activity controls the earthquake hazard level that, due to the occurrence of secondary and tertiary effects, has also direct implications on the hazard related to mass movements (e.g., landslides and naturally dammed lake outbursts). Note that in Central Asia landslides, mudslides and debris flows are causing an extensive number of casualties every year. Climatically, this region is characterized by strong rainfall gradient contrasts, due to the diversity of climate and vegetation zones. The region is drained by large, partly snow- and glacier-fed mountain rivers, that cross or terminate in arid forelands. Central Asian countries are therefore affected by a significant river flood hazard, mainly in spring and summer seasons. The challenge posed by this combination of different hazards can only be tackled considering a multi-hazard approach harmonized among the different countries, in agreement with the requirements of the Sendai Framework for Disaster Risk Reduction, approved at the third UN World Conference on Disaster Risk Reduction in 2015.

This report describes the methodology and results of Task 3, *Fluvial and Pluvial Flood Hazard Assessment*, within the "***Regionally consistent risk assessment for earthquakes and floods and selective landslide scenario analysis for strengthening financial resilience and accelerating risk reduction in Central Asia***" project. Flood hazard assessment was carried out by means of a physically based numerical modelling toolset and a stochastic catalogue of flood footprints. In particular, the report includes the following topics:

- In Section 1, a brief introduction, describing the scope of Task 3, state of the art of flood hazard assessment and summary of our methodology.
- Section 2 describes the available global and local datasets
- The following Sections are dedicated to the presentation and validation of the modeling toolset, which include the hydrological model TOPKAPI (Section 3), Extreme Value Analysis (Section 4) and the hydraulic model CA2D (Section 5)
- In Section 6 we present three realistic flood scenarios that have not happened in the past but may affect the region in the future.
- The generation of the stochastic catalogue is described in Section 7
- Section 8 focuses on the climate change scenario
- Section 9 is dedicated to pluvial flood modeling
- Finally, in Section 10 we present our work's strengths and limitations as well as recommendations for future users
- Annexes 1, 2, 3 and 4 contain all flood hazard maps (fluvial undefended, fluvial defended, fluvial future scenario, pluvial)

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1 Introduction

1.1 Scope of Task 3 – Flood hazard assessment

Central Asia is highly exposed and vulnerable to a broad range of natural hazards which frequently result in economic and human losses. Flood hazard is significant in the region, with floods being the most frequent natural disaster in the period 1988-2007 according to a recent analysis provided by the Central Asia and Caucasus Disaster Risk Management Initiative (CAC DRMI, 2009). In the same period, floods were second for number of deaths and amount of population affected (1,512 and 19% respectively). Despite the aridity of large areas in some of the target countries, natural phenomena linked to extreme precipitation can cause billions of dollars of damages every year: collectively, floods inflict the second highest overall economic losses ($52 million), surpassed only by earthquake (an annual average of $186 million). At the local level (e.g., in Tajikistan), flood is sometimes the dominant risk in terms of economic losses (World Bank et al., 2012). Considering the deteriorated protection infrastructure and vulnerabilities in several sectors, floods cause considerable damage to housing, infrastructure, and agriculture (Libert, 2008).

A robust estimation of possible loss due to flood is necessary for developing and implementing successful regional DRM strategies as advocated for by the past World Conferences for Disaster Risk Reduction (UNDRR, 2019). In order to support the development of regional DRFI solutions, and to inform DRM/DRR activities in the region, in this study we assess fluvial and pluvial flood hazard to be used as input to characterize flood risk in Central Asia. We adopted a hybrid stochastic and physically based approach, which takes into account the scale of the phenomenon, the data available (both input data and calibration/validation data) and potential future modification to the hazard triggered by climate change.

For the purpose of this study, we distinguish two common types of floods: fluvial flood, which occurs when excessive rainfall over an extended period of time causes a river to exceed its capacity, and pluvial flood, which occurs when heavy rainfall creates a flood event unrelated to an overflowing water body. The final deliverables are:

a) Fluvial and pluvial flood hazard maps for current climate over the entire Central Asia area including Kazakhstan, Kyrgyz Republic, Uzbekistan, Tajikistan, and Turkmenistan for selected return periods (5, 10, 20, 50, 100, 200, 500, 1000 years); fluvial flood hazard for current climate is assessed in both undefended and defended scenarios and a variability range is also provided for the undefended case
b) A hazard-consistent stochastic catalogue of fluvial flood footprints, which is essential for characterizing risk and supporting DRM strategies
c) Fluvial flood hazard maps for the same selected return periods obtained taking into account the impact of climate change for a 2080 scenario. Fluvial flood hazard for current climate is assessed in both undefended and defended scenarios and a variability range is also provided for the undefended case
d) Hazard curves at five selected locations
e) Flood footprint for one historical scenario used to validate the CA2D hydraulic model
f) Flood footprint for three realistic scenarios across the region representing potential flood events with 100-year return period that may occur in populated and flood-prone areas.

The proposed modelling approach, which leverages on the use of reanalysis data available in near real time (e.g., ERA5-Land) and on a hydrological/hydraulic modelling chain calibrated using historical observed data, is potentially suitable for the implementation of Disaster Risk Financing applications based on parametric triggers (e.g., CCRIF for the Caribbean and Central America regions, ARC for the continent of Africa).

The development of a regional model cannot be done without the contribution of experts from the local scientific community. Partnership with local governmental institutions and authorities is also an essential step to facilitate model acceptance and for potential integration with national models. Following this concept, the consortium has engaged with the local communities for building and extending awareness of flood hazard and for enhancing the technical capacity of local experts in the use of open tools and resources (see Table 1 for the complete list of involved scientific institutions from each partner country).

Table 1. List of partner countries of the consortium and associated scientific institutions

Country	Main Scientific Institution	Local Representative and support team
Kazakhstan	IS under MoES of RoK - Institute of Seismology	Dr. Natalya Silacheva Dr. Zhanar Raimbekova
Kyrgyz Republic	ISNASKR - Institute of Seismology of Kyrgyz Republic	Prof. Kanatbek Abdrakhmatov Dr. Anna Berezina Dr. Ruslan Umaraliev
Tajikistan	IWPHE - Institute of Water Problems, Hydropower Engineering and Ecology	Prof. Zainalobudin Kobuliev Dr. Sitora Mirzokhonova Dr. Mirzo Saidov
Uzbekistan	ISASUz - Institute of Seismology Uzbekistan	Prof. Vakhitkhan Ismailov Dr. Vakhob Rafikov Dr. Nodirbek Tolibov Dr. Nadirzhan Mamadaliev
Turkmenistan	Various individual consultants	Dr. Japar Karaev Dr. Vladimir Belikov

1.2 Previous regional and global hazard studies

Regional flood hazard assessment for Central Asia is usually included in more comprehensive risk studies. A study on flood hazard of the territories in the Esil Water Management Basin in Kazakhstan was conducted in 2019 (Plekhanov and Medeu, 2019). In Kyrgyz Republic, a catalogue of historical floods and hazard maps is managed by the Ministry of Emergency Situations of the Kyrgyz Republic (MES). At the global scale, there are various flood hazard assessment models that provide hazard maps at various resolutions and considering different return periods. Some examples are the Catchment-Based Macro-scale Floodplain (CaMa-Flood) model (Yamazaki et al., 2011), the Centro Internazionale in Monitoraggio Ambientale and United Nations Environment Program (CIMA-UNEP) model (Rudari et al., 2015), the European Centre for Medium-Range Weather Forecasts (ECMWF) model (Pappenberger et al., 2012), the Global Flood Risk with Image Scenarios (GLOFRIS) model (Ward et al., 2013) which is implemented in the Aqueduct Floods platform (Ward, P. J. et al., 2020), the Joint Research Centre (JRC) model (Dottori et al., 2016) based on the Copernicus Global Flood Awareness System (GloFAS) (Alfieri et al., 2013), and the Fathom Global Ltd model (Sampson et al., 2015). For these models, the corresponding flood

hazard maps resolution at the equator vary between ~90 m (Fathom Global Ltd, CIMA-UNEP), ~540 m (CaMa-Flood, ECMWF), and ~900 m (GLOFRIS, JRC GloFAS) (Bernhofen et al., 2018). However, these models only use global data and adopt a global approach that was not specifically tailored or validated for Central Asia.

The results presented herein are intended to meet the need for a homogeneous national and regional flood hazard assessment in Central Asia, combining in a unified regional framework both global and unprecedented local data at a high resolution (~90 m).

1.3 Methodology overview

In this study, the flood hazard of five Central Asian countries (Kazakhstan, Kyrgyz Republic, Tajikistan, Turkmenistan, and Uzbekistan) is assessed by means of a physically based numerical modelling toolset and a stochastic catalogue of flood footprints (Figure 1). The numerical modelling toolset is composed of two elements: the hydrological model (TOPKAPI) and the hydraulic model (CA2D). A statistical correction of extreme values and their regionalization are also performed based on the availability of a historical records of daily and annual maxima discharges.

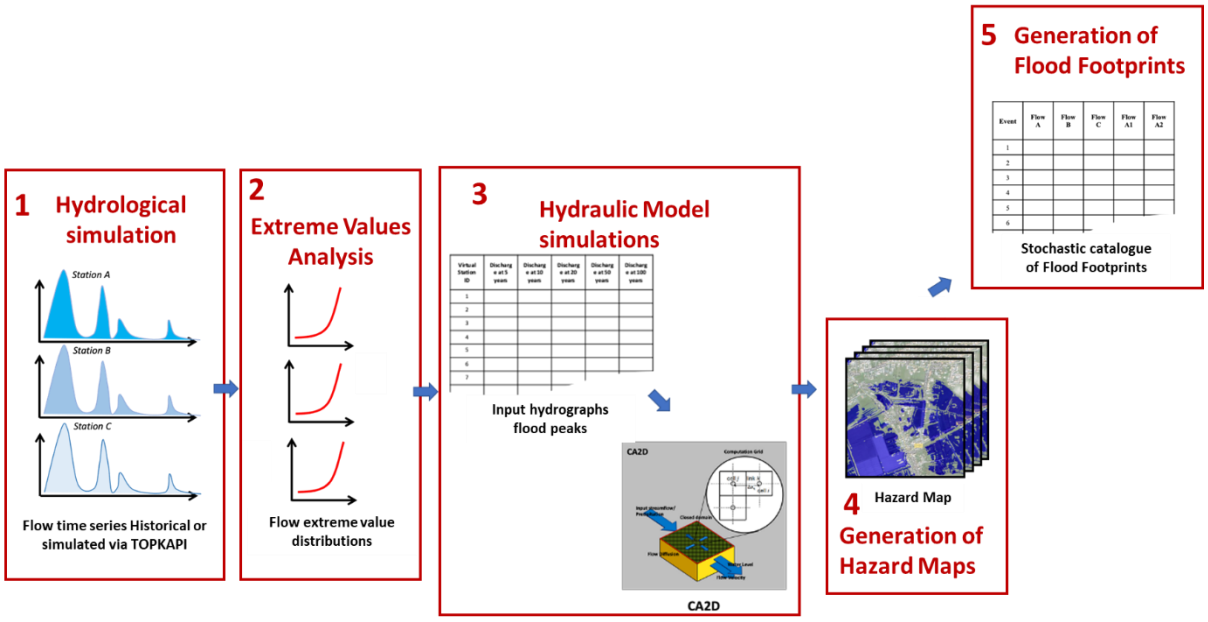


Figure 1. Fluvial flood hazard assessment: schematic representation of methodology

The TOPKAPI (TOPographic Kinematic APproximation and Integration) model is a fully-distributed physically-based hydrological model that can provide high resolution information on the hydrological state of a catchment (Ciarapica and Todini, 2002). The TOPKAPI model requires as input both precipitation and temperature meteorological data, plus a description of the soil characteristics that can be derived from the land use (to derive crop factors and surface roughness) and soil type maps (to derive soil permeability and depth).

CA2D (Dottori and Todini, 2011) is a full physically-based hydraulic model specifically designed for high-performance computing applications, based on the cellular automata (CA) approach and the diffusive wave equations, to simulate flood inundation events involving wide areas. The model

is based on the state-of-the-art of large-scale hydraulic modelling and has been tested extensively on several case studies. The CA2D model has an internal preprocessor that allow the user to provide as input only the Digital Elevation Model and the surface roughness map. The network (comprising nodes and links) is automatically generated, and specific conditions (such as flood protections) can be included where present. In addition, input meteorological data must be provided in the form of hydrographs at specific points and/or of rainfall maps.

The two models are linked together as shown in Figure 1: (1) the TOPKAPI hydrological simulation results informs the (2) Extreme Value Analysis and regionalization, which produces a stochastic description of streamflow from which peak flows values are extracted (1-in-5, 10, 20, 50, 100, 200, 500, 1000, 2000 years floods) and used as input for the (3) CA2D hydraulic model, which simulates the channels routing as well as the 2D inundation processes to derive the (4) hazard maps. Finally, the hazard-consistent stochastic catalogue of flood footprints for the region is produced.

While hazard maps provide the probability and depth of inundation at a given location, they are unable to describe the likelihood of concurrent flooding across multiple sites, limiting their capability of assessing risk over the full range of plausible scenarios, including the most extreme ones which are the very thing insurers must quantify for pricing and liquidity. A flood hazard catalogue serves this purpose, by providing a stochastic ensemble of 10,000 years of all hypothetical floods, with related annual frequency of occurrence, that may occur in the region of interest. To ensure spatial coherence in the stochastic catalogue, the spatial correlation of the river flow at each gauge/station is determined by computing a cross correlation matrix on all the available (observed/simulated) flow time series. The procedure is detailed in Section 7.

The Extreme Value Analysis and regionalization process consists of two steps: a) a correction of the TOPKAPI simulated discharge based on the historical record of observed data, which allows for better representing the extreme values simulated during the period 1981-2020, and b) the estimation of General Extreme Value distribution for several locations along the drainage network to derive the peak flows with different return periods. Additional details will be given in Sections 3 to 8.

Pluvial flood hazard is estimated applying a similar methodology (Figure 2), where, to reduce the computational effort, a geomorphological technique is combined to the CA2D hydraulic simulations. First, annual maxima precipitation for different durations derived from ERA5-Land gridded data, which are available from 1981 to present, are corrected based on their comparison with the station data from the KNMI Climate Explorer dataset. This correction was necessary to take into account the unavoidable discrepancy between point data and gridded data. In fact, the latter represent the average precipitation over an area of about 100 km^2 and does not well describe the extremes that may occur in smaller sectors. The corrected ERA5-Land annual maxima precipitation is eventually used to derive Intensity-Duration-Frequency curves, from which the precipitation at different return periods is extracted to feed the CA2D hydraulic model. The CA2D model is applied to several areas and the output are used to calibrate the geomorphological technique based on the Compound Topographic Index to extend the simulation to the entire region. Additional details will be given in Section 9.

We did not include the generation of a pluvial flood hazard stochastic catalogue because pluvial flood is a much more local phenomena and, on such a large area, it requires an impractical

computational effort to effectively run stochastic/climatic models. Moreover, the level of uncertainty would be too large to produce reliable results to be used for risk estimation.

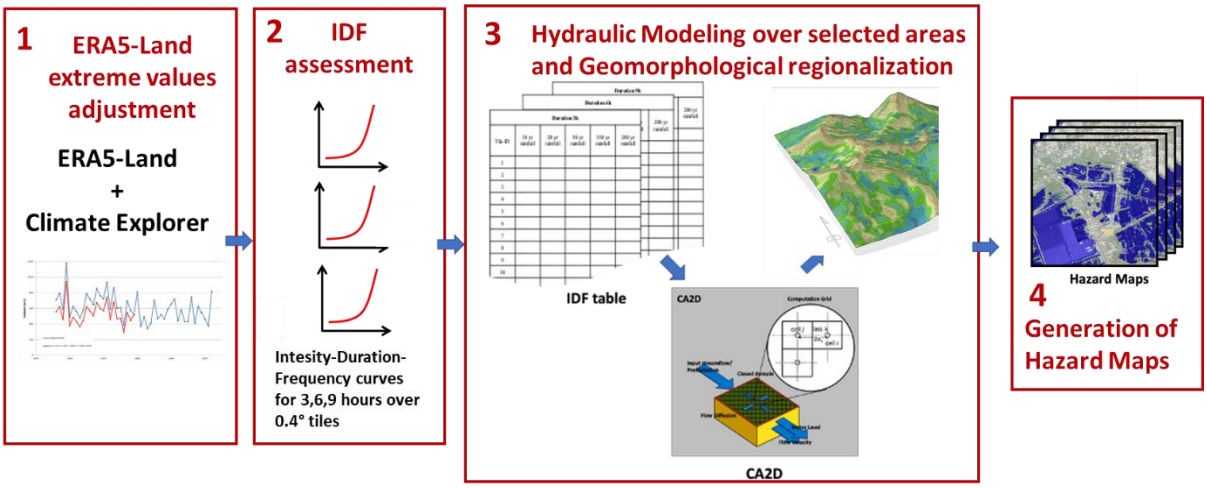


Figure 2. Pluvial flood hazard assessment: schematic representation of methodology

2 Data Collection

Both the hydrological and the hydraulic model require daily or sub-daily hydrometeorological data as well as spatially distributed information such as soil characteristics, land use, elevation, and the presence of hydraulic structures and reservoirs. With the aim of obtaining a homogeneous and regionally consistent hazard assessment, we integrated the ground local data, available from historical observations and previous national and regional studies, with spatially distributed global datasets.

Specifically, the TOPKAPI hydrological model requires daily or sub-daily precipitation and temperature records as input data and discharge time series at daily/sub daily resolution for calibration and validation. Discharge time series are also used to characterize the statistical behavior of extreme flows through the analysis of annual maxima, which is essential to produce the inputs for CA2D hydraulic simulations. CA2D also requires a detailed digital elevation model to define the boundary conditions of the domain. Both TOPKAPI and CA2D require spatially distributed information for the characterization of the river catchments and flow behavior (infiltration and roughness). Data on defensive protections and the presence of regulated/unregulated reservoirs and dams is also critical for accurately assess flood hazard, as a misrepresentation of these structures can lead to over/underestimation of flow and water depth magnitude and frequency. Unfortunately, information regarding hydraulic defenses is rare and very difficult to obtain because a comprehensive database is usually not available at country level, and it is considered classified.

2.1 Local data

We held multiple individual meetings with our partners located in the targeted countries to illustrate our methodology and the data needed to accomplish the flood hazard assessment. We also clarified the requested data temporal resolution and format. Specifically, we focused our data requests on:

1. Historical records of discharge data at daily or sub-daily temporal resolution. Alternatively, records of monthly or annual maximum discharge.

2. Information on fluvial hydraulic protections including the poly-lines of the protections and potential information on height and protection level.

3. Location of lakes and reservoirs potentially including the polygons describing their spatial extent and associated with information on volume, operational rules, relation volume-level, relation level-discharge.

There was a large variability in the type and quality of the data provided. Specifically, for most countries, obtaining daily discharge data was a long process that required several external meetings and official requests to the partner Countries' Ministries. Another common issue was digitization, as data from local partners were typically on paper form. Finally, all local partners made clear that a comprehensive database of hydraulic protection data has never been compiled at country level in any of the targeted countries and this type of information is considered highly classified and confidential. Unfortunately, it would be extremely difficult to obtain it even at a low level of detail. Hence, all the local institutions that we contacted stressed the impossibility to disseminate such confidential information.

Table 2 shows a complete inventory of the data requested/obtained from the local partners.

Table 2. Local data inventory

Country	Daily Discharge	Annual maximum discharge	Hydraulic Protection	Reservoirs
Kazakhstan	7 stations (records of variable lengths between 2001 and 2015)	120 stations (records of variable lengths between 1910 and 2018)	Location and length of some riverbanks hydraulic structures on the Sir Darya River	Volume and year of construction of main reservoirs
Kyrgyz Republic	No data obtained because cost was too high compared to the benefit	65 stations (records of variable length between 1930 and 2018). Some of these data were purchased from KyrgyzHydroMet	Record of hydraulic protection works from 2018 at the Oblast level	Data on reservoirs' volume and construction year for 5 main reservoirs
Tajikistan	No data obtained because cost was too high compared to the benefit	14 stations (variable lengths), purchased from TajykHydroMet	No data	Volume and construction year for 13 reservoirs
Uzbekistan	2 stations (2015-2019)	46 stations (2005-2019)	No data	Volume and year of construction of main reservoirs
Turkmenistan	6 stations (2015-2020)	11 stations, variable record between 1936-2020 (monthly maxima available)	No data	Volume and year of construction of main reservoirs

Even if the number of available flow gauges seems relatively small for such a large area, it is important to note how the spatial distribution of these stations conveniently covers the populated areas i.e., those where we expect to find exposed assets. Figure 3 and Figure 4 show the gauging stations coverage of populated areas (in blue, population density > 1/km^2) including both local and global datasets.

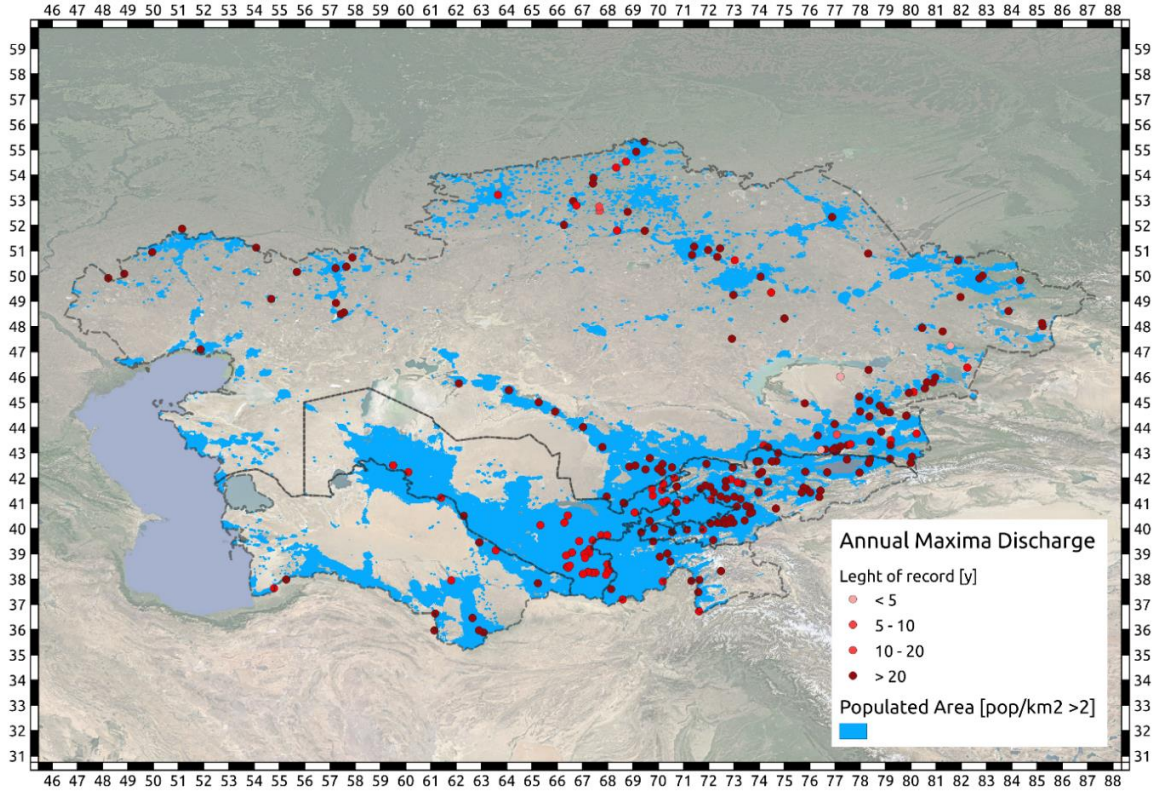


Figure 3. Annual maxima flow gauges coverage on populated areas, provided by local Institutes

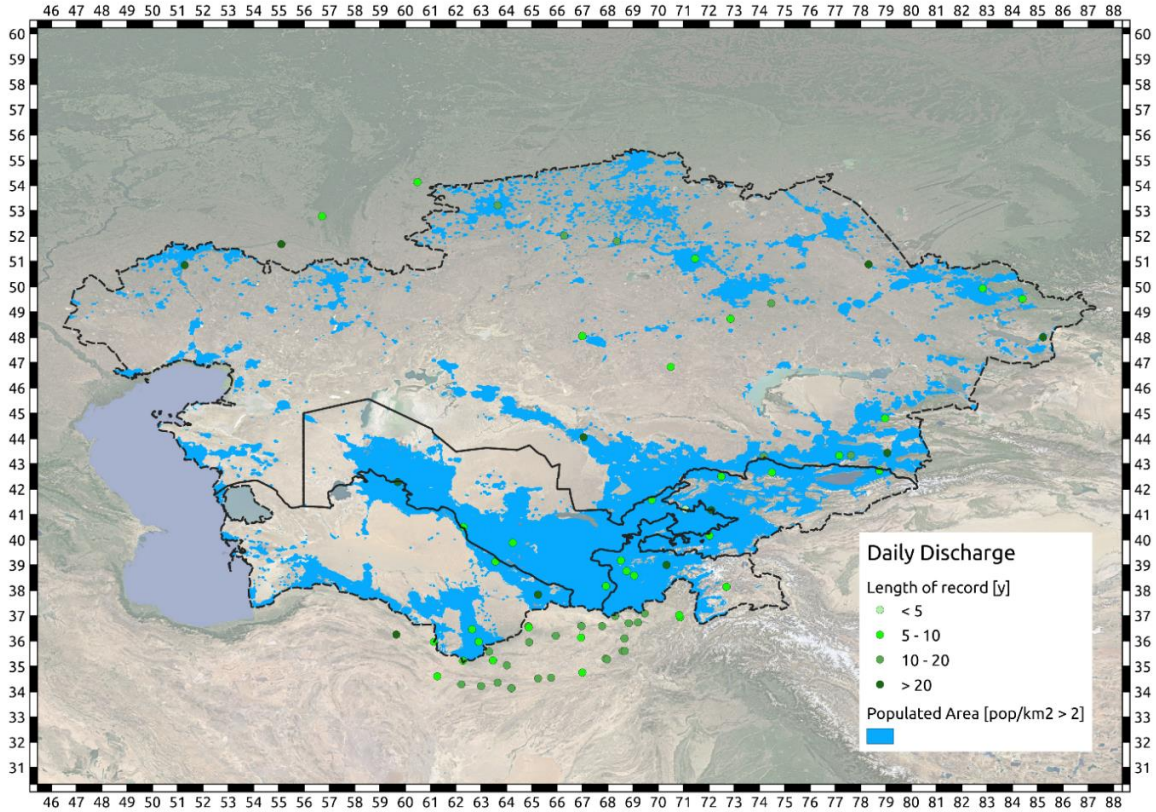


Figure 4. Daily flow gauges coverage on populated areas, provided by local Institutes (inside of the country borders) and GRDC (inside and outside of the Countries' borders)

2.2 Global data

The following global datasets were used to integrate and/or replace missing consistent local information, a summary is provided in Table 3:

- *Elevation:* MERIT DEM (Karnieli et al., 2010) together with MERIT Hydro (Yamazaki et al., 2019) were used. MERIT Hydro is a new global flow direction map at 3 arc-second resolution (~90 m at the equator) derived from the latest elevation data (MERIT DEM) and water body datasets G1WBM, GSWO (which stand for Global 1 arc-second Water Body Map and Global Surface Water Occurrence, respectively, and were developed by the same authors as MERIT) as well as OpenStreetMap (Coast, 2005). MERIT Hydro, which was produced at the University of Tokyo, uses a new algorithm to extract river networks near-automatically by separating actual inland basins from dummy depressions caused by the errors in input elevation data. MERIT Hydro improves on existing global hydrography datasets in terms of spatial coverage and representation of small streams, mainly due to increased availability of high-quality baseline geospatial datasets.
- *Soil characteristics:* the FAO Harmonized World Soil Database was used, a 30 arc-second raster database with over 15,000 different soil mapping units that combines existing regional and national updates of soil information worldwide (Nachtergaele et al., 2012).
- *Land use:* GlobeLand30, a 30-m resolution, 10 classes Global Land Cover dataset developed by the National Geomatics Center of China (NGCC) and provided with open access to the UN (UN-ESCAP Statistics Division; UN_ESCAP ICT and Disaster Risk and Reduction Division, 2015) was used.
- *Observed discharge:* daily streamflow records from the Global Runoff Data Centre (GRDC) were used.
- *Precipitation and Temperature:* the ECMWF ERA5-Land (Muñoz-Sabater et al., 2021) and the KNMI Climate Explorer (Trouet and Van Oldenborgh, 2013) datasets were used. ERA5-Land provides hourly 0.1°x0.1° estimates for both precipitation and temperature combining model data with observations from across the world into a globally complete and consistent dataset, ERA5 is updated daily with a latency of about 5 days. The data are available from January 1981 to present. We used observed data from the KNMI Climate Explorer to assess and correct the ERA5-Land extreme precipitation estimates due to the discrepancy between point station data and grid averaged data.
- *Reservoirs and Dams:* we used the Global Reservoir and Dam Database (GRanD), which provides information on dams' location, storage and first year of operation for over 6,862 dams and reservoirs worldwide (The Global Water System Project, 2011).
- *Hydraulic defenses:* the mostly local and qualitative information on hydraulic protection collected from the local partners were integrated into our strategy to define hydraulic defenses in the area that combines the FLOPROS database, providing hydraulic protection standards (Scussolini et al., 2016), with the WorldPOP and the Landsat HBASE databases on urban areas and population density (Tatem, 2017; Wang et al., 2017).

Table 3. Global data

Data Type	Source	Use within the project
Digital Elevation Model	MERIT DEM MERIT Hydro (See Figure 5)	Input to TOPKAPI, CA2D. Input to derive additional data products
Soil Type	FAO Harmonized World Soil Database (See Figure 6)	TOPKAPI parametrization
Land Use	GlobeLand30 (see Figure 7)	TOPKAPI and CA2D parametrization.
Observed Discharge records	Global Runoff Data Centre (GRDC)	TOPKAPI calibration and extreme values distributions
Precipitation	ERA5-Land: 1981-2020 KNMI Climate Explorer	Input to TOPKAPI and CA2D pluvial simulations
Temperature	ERA5-Land: 1981-2020 (see Figure 8 and Figure 9)	Input to TOPKAPI
Reservoirs and Dams	Global Reservoir and Dam Database (GRanD)	Extreme Values Analysis and Regionalization
Hydraulic defenses	FLOPROS database WorldPOP HBASE	Defended hazard maps

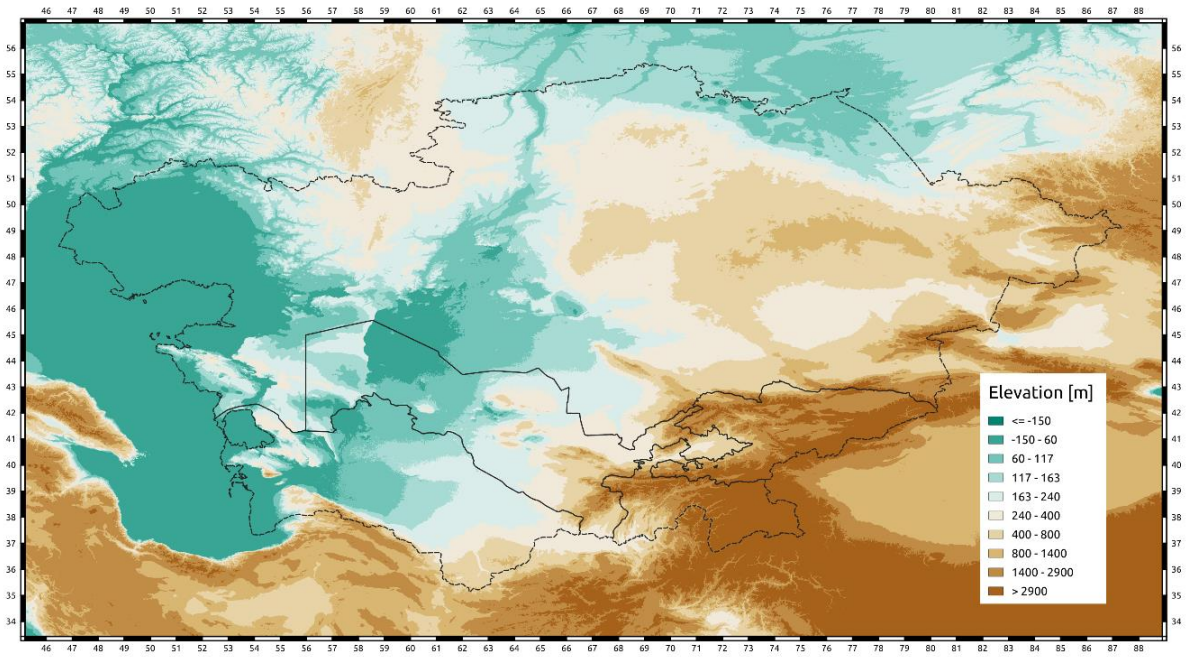


Figure 5. MERIT Digital Elevation Model over Central Asia

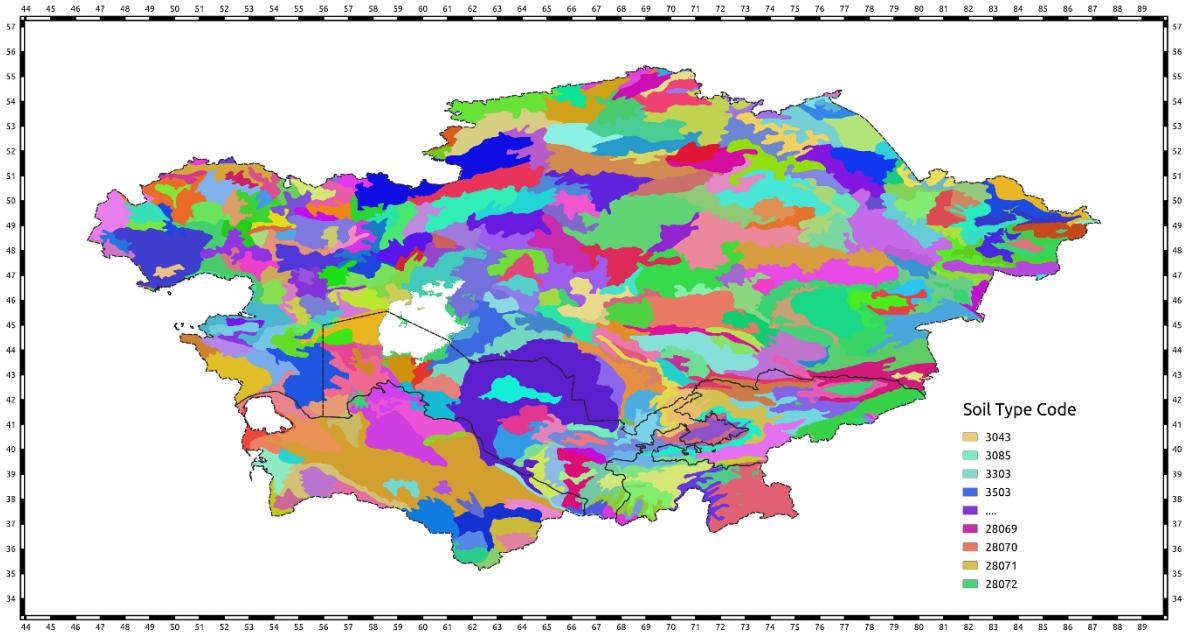


Figure 6. FAO Harmonized World Soil Database over Central Asia, each color represents a typology of soil according to the FAO soil classification

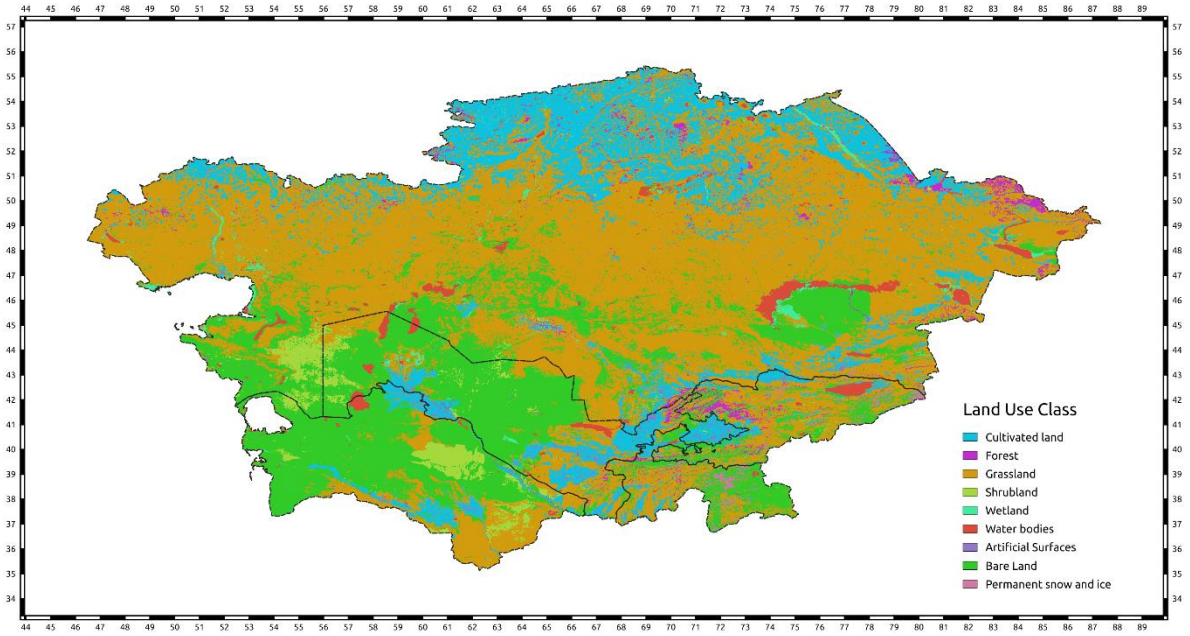


Figure 7. GlobeLand30 land use over Central Asia

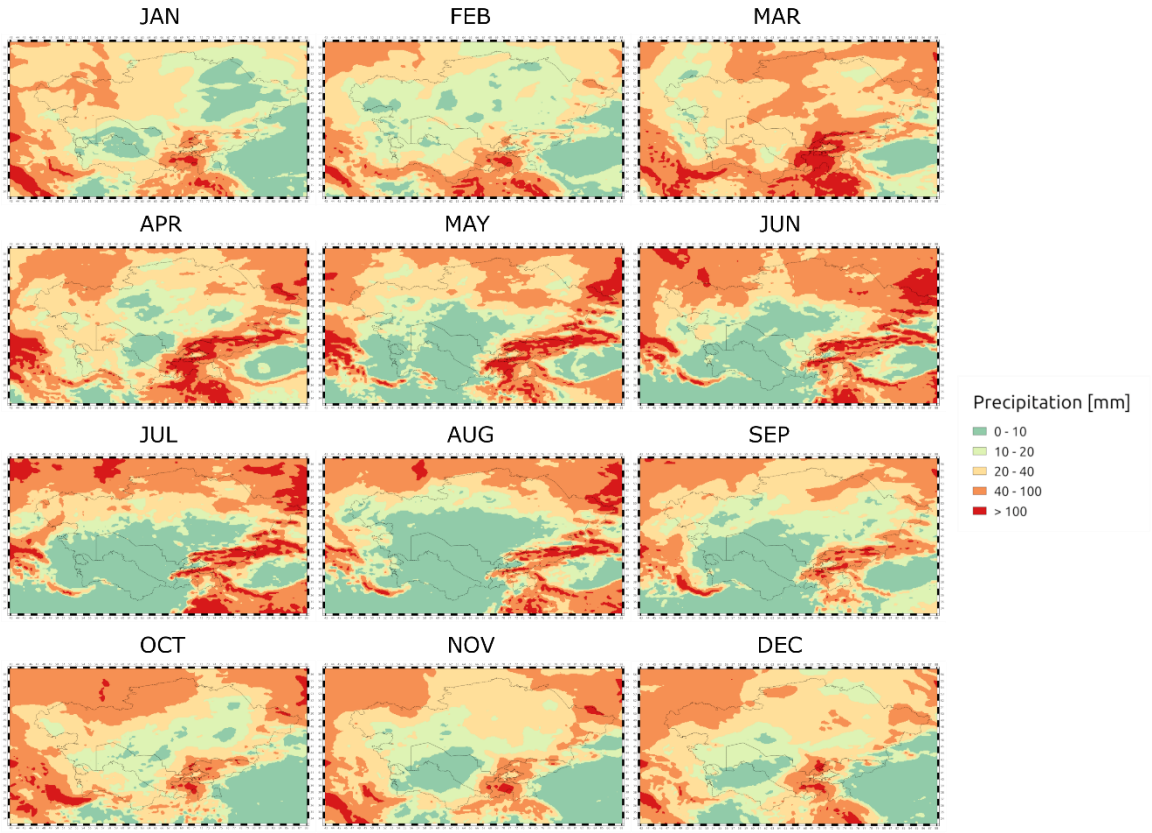


Figure 8. ERA5-Land monthly average precipitation during the period 1981-2020

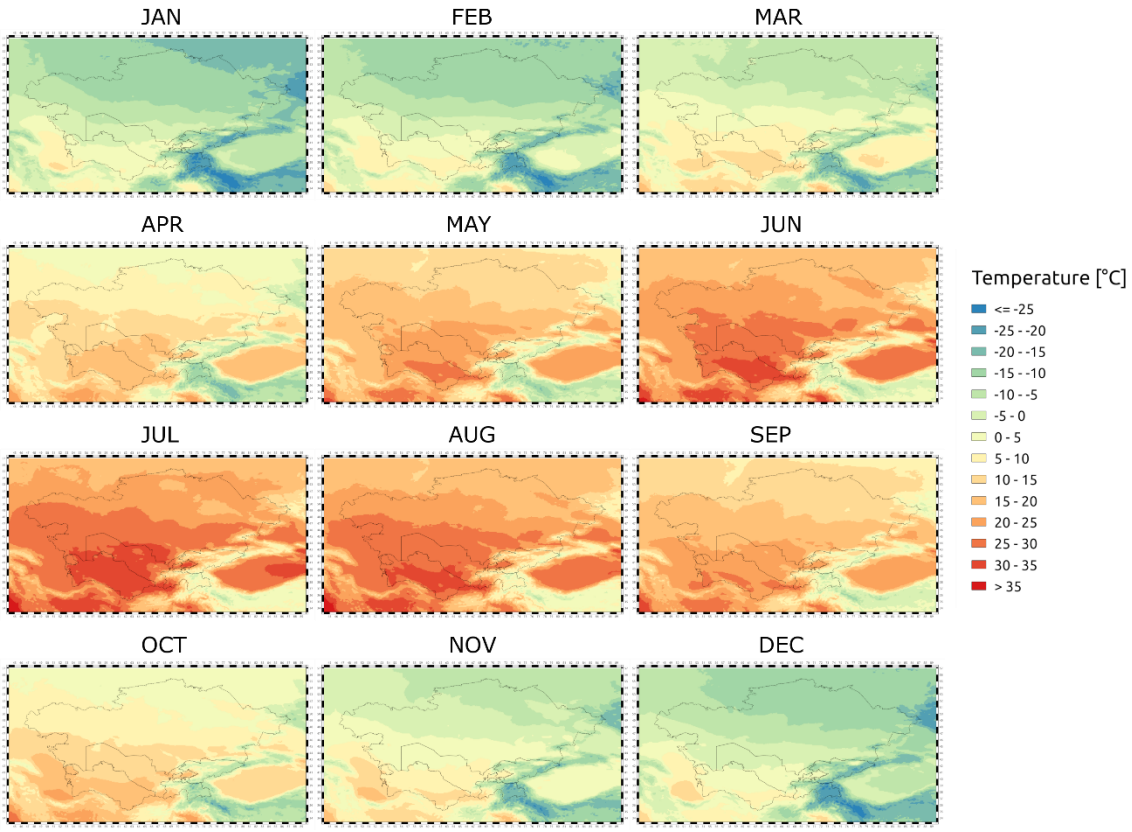


Figure 9. ERA5-Land monthly average temperature during the period 1981-2020

3 Hydrological Modeling

The TOPKAPI (Ciarapica and Todini, 2002) hydrological model is a well-established, cutting-edge tool derived upon the assumption that the horizontal flow at a point in the soil and over the surface can be approximated by means of a kinematic wave model. The model was first developed around 20 years ago, and then constantly updated to keep up with the latest advances in the field. The consortium holds an internal version of the TOPKAPI model, which was set up at the catchment scale in order to optimize its performances in terms of computational times over the entire Central Asia region. The model was run with a hourly time step and 1-km spatial resolution. The model was applied to all major catchments in the countries of study, including transboundary catchments, also covering areas in China, Afghanistan and Russia that contribute to the runoff of the rivers crossing the targeted countries.

3.1 Model Setup

TOPKAPI is a fully distributed physically based hydrological model that combines the kinematic approach with the topography of the basin described by means of a lattice of square cells. Each cell represents a computational node for the physical characteristics of the model, namely the mass balance and the momentum balance. TOPKAPI takes precipitation and temperature meteorological data as inputs and evaluates the flow paths from the DEM. Crop factors, surface roughness, soil permeability and depth are derived from land-use and soil-type maps and are utilized to simulate infiltration and flow behavior.

The TOPKAPI model includes a specific module for managing reservoirs and regulated lakes. However, it needs to be calibrated with specific information on the operation strategies and/or volume/level curves for all reservoirs. Since such data were not generally available for the simulation domain, we did not include the lake module within the TOPKAPI configuration but took eventually into consideration the effect of reservoirs by correcting the resulting flow curves in the river branches downstream dams using the procedure explained in Section 4.1.2.

Figure 10 schematically represents the structure of the TOPKAPI model and highlights the main hydrological processes that are simulated: snowmelt, evapotranspiration, three levels of horizontal flow (overland, sub-surface and channel), infiltration, and percolation.

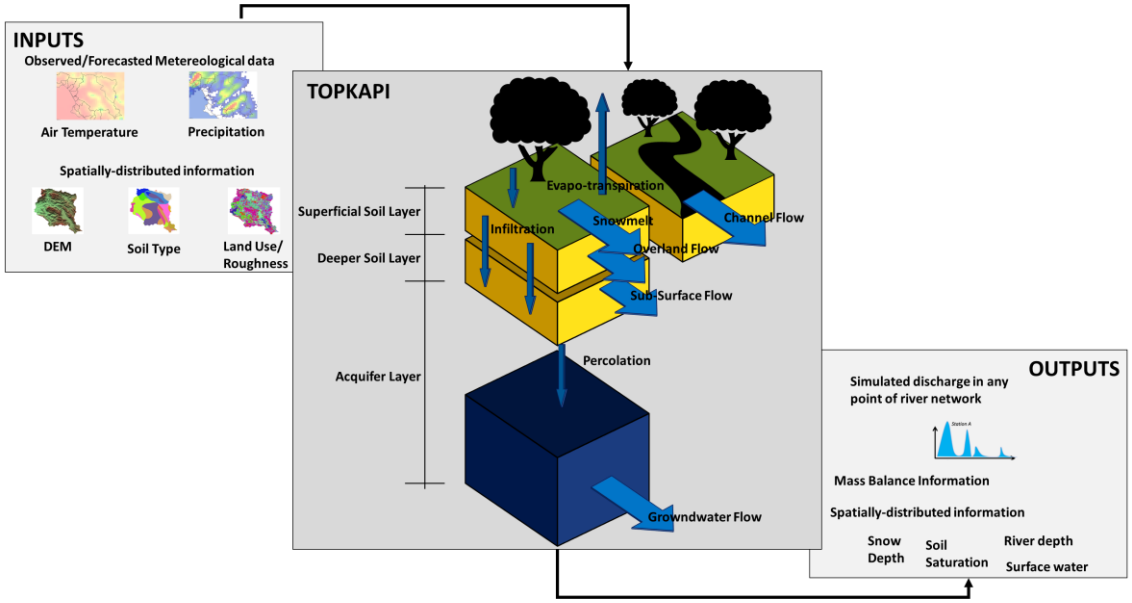


Figure 10. Conceptual scheme of the TOPKAPI hydrological model. The non-linear reservoirs that describe the four levels of flow are shown: overland, sub-surface from superficial layer (~ 10-80cm), sub-surface from deeper layer (~ 1-3m), and channel.

3.1.1 Data processing

TOPKAPI uses a regular grid domain. For this reason, we decided to adopt the Lambert Azimuthal Equal Area (LAEA) projection for TOPKAPI data inputs.

MERIT DEM and MERIT Hydro were used to produce the following maps:

- catchments boundaries at 90 m (see Figure 11);
- main rivers network at 90 m;
- slope along the flow direction at 90 m.

Data was then reprojected to LAEA at 1 km for all catchments (see an example for the Ural River basin in Figure 12). During this process we aimed at keeping consistency between the original 90 m dataset and the reprojected datasets at 1 km in terms of river network identification and catchment area. In fact, the upscaling process to create the TOPKAPI configurations was forced to match, when possible, the hydrological network and the catchment boundaries that were obtained using the original 90m dataset from MERIT Hydro. Similarly, ERA5-Land precipitation and temperature data were extracted and reprojected as hourly data over the domain with a 10 km spatial resolution.

Soil Type and Land Use maps were also reprojected to LAEA at 1 km and cropped over the domain. Based on literature and our experience with previous applications of the TOPKAPI model, initial values for the physical model parameters were associated and derived for each class of soil type and land use included in the domain.

Finally, discharge data over the region from both local sources and GRDC were processed to be used for the model calibration and validation.

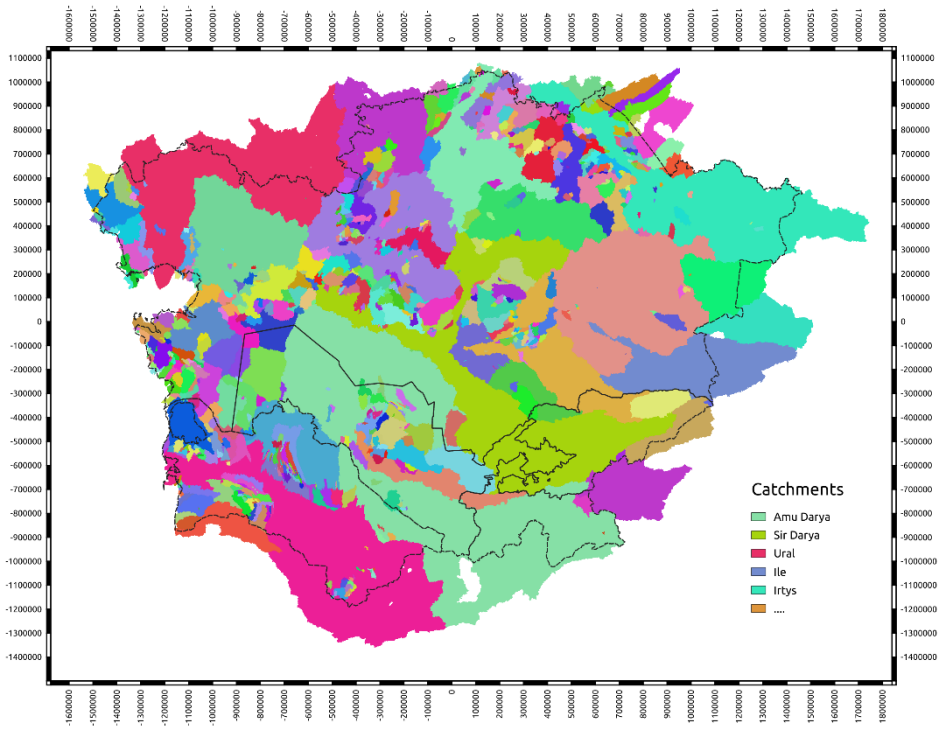


Figure 11. Catchments boundaries at 90m derived from MERIT Hydro (LAEA coordinate system)

3.1.2 TOPKAPI model simulations' setup

Once all the input data have been processed, the final TOPKAPI model setup at 1 km spatial resolution included for each catchment:

- Catchment boundary derived from the original 90m MERIT Hydro DEM
- Digital Elevation Model derived from the original 90m MERIT Hydro DEM and adjusted to be hydrologically consistent (single outlet, no sinks, unique drainage network)
- River network derived from the DEM mentioned in the previous point
- Soil type maps derived from the FAO HSWD dataset
- Land use map derived from the GlobeLand30 dataset
- Hourly precipitation data from January 1981 to December 2020 interpolated from the original 0.1° ERA5-Land dataset
- Hourly temperature data from January 1981 to December 2020 interpolated from the original 0.1° ERA5-Land dataset
- Daily observed discharge record for the stations showed in Figure 4 derived from both local institutes and GRDC; the GRDC stations covered mainly the period 1981-1990, while the data from local institutes had a variable length included between 2000 and 2020
- Annual Maxima discharge record for the stations shown in Figure 3 spanning different periods between 1981 and 2018

For each catchment, the TOPKAPI model has been simulated at hourly time resolution for a period of 40 years, from January 1st 1981 to December 31st 2020. The model simulations have been initiated with average soil saturation and river depth conditions and the first year of simulation has

been used as warm-up period to reach realistic conditions. Therefore, the period January 1st 1981 to December 31st 1981 has not been considered for calibration purposes nor for the extreme value analysis (Section 4).

The main model output consisted in hourly simulated discharges at several location of the river network across the entire region. The output corresponding to locations where observations were available have been used to perform a trial-and-error calibration that could reasonably reproduce the overall behavior of each catchment (Section 3.2).

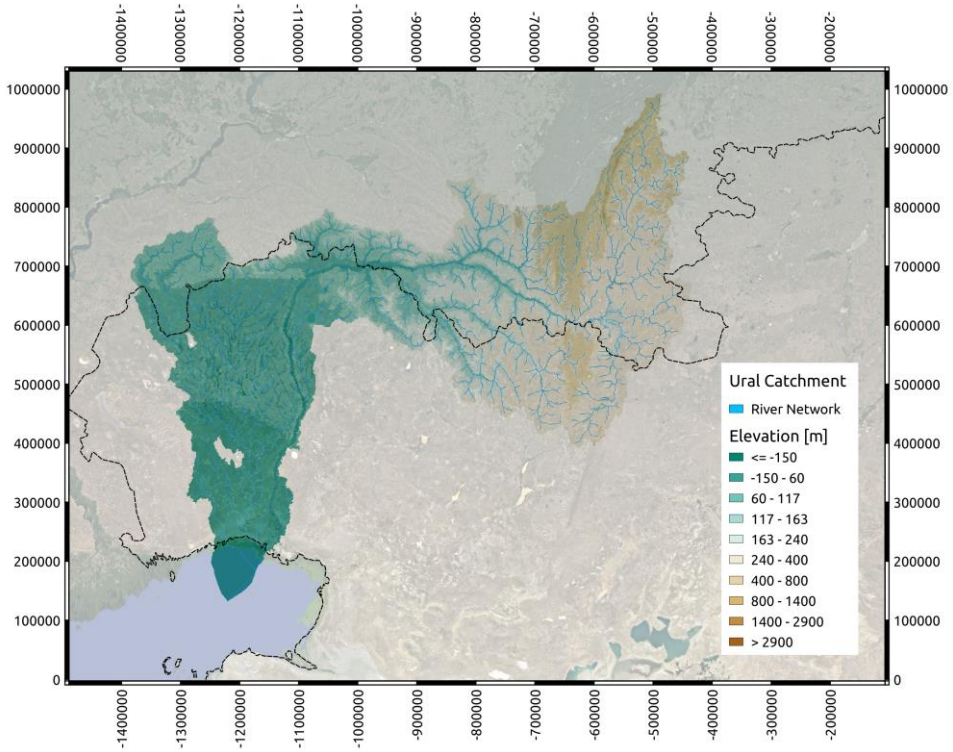


Figure 12. Example of TOPKAPI configuration at 1 km spatial resolution for the Ural River catchment

3.2 Calibration and Validation Results

The TOPKAPI model was calibrated through a trial-and-error procedure adapting the initial model parameters in order to match the available observed discharge. We mainly based the model calibration on the historical daily data, but also used the annual maxima for the areas where daily data were not available. Given that the objective of the hydrological simulations is the derivation of the extreme discharge value distributions over several river sections across the region to eventually feed the hydraulic model to derive the flood footprint at different return periods, the calibration process focused on reproducing the flow peaks. Since a physically based model reproduces better the flow peaks if all the other hydrological components are well represented, we made sure that the main hydrological processes were also correctly reproduced, with particular attention to the snow accumulation/melting component, which is the driver of most of the floods affecting the region. The calibration has been performed independently for each catchment where historical data were available. Given the distributed and physically based nature of the TOPKAPI hydrological model, the calibration process was not based on an automatic procedure but on the

use of reasonable values of the physical parameters. In fact, the calibration process was based on an iterative procedure through the following steps:

- Adjustment of model parameters (mainly soil conductivities, soil depth, snow parameters and crop factors) to match the observed data of upstream stations where daily, or at least annual maximum, historical observations were available. Expert judgment, analysis of water balance among the different hydrological component (e.g., precipitation, snow melting, evapotranspiration and river volume) and evaluation indexes such as Nash-Sutcliffe, bias and correlation were used to identify the best set of parameters.
- Validation of calibrated parameters against the observed discharge at downstream stations based on expert judgment and evaluation indexes
- Adjustment of model parameters based on the result of the validation mentioned at the previous step
- Validation of adjusted parameters against the observed discharge at upstream stations used in the first step of the loop

This procedure allows the identification of model parameters values that provide reasonable outputs for the entire catchments, avoiding under- or over-fitting any of the available historical records. During the calibration process, the main limitations of the hydrological model application were taken into account:

- The absence of lake/reservoirs modeling was taken into account allowing an overestimation of the flood peaks and underestimation of the low flows for the stations located downstream the main reservoirs. In fact, the reservoirs cause the lamination of the flood wave storing water from the flood peak and releasing it during the dry period. This behavior was not explicitly modeled due to lack of information on the reservoir management strategies, therefore we expected the hydrological model not to match precisely the observed hydrograph for the stations downstream reservoirs (e.g., the Turkmenabat, Atamurat, Semiyarskoje and Chirchik stations in Figure 13)
- The inability to reproduce the effect of alluvial plain, which modify the flood waves following a similar pattern to that described for the reservoirs, namely flattening the flood peak. Therefore, also in stations locate in alluvial plains we allowed the overestimation of the peaks and the underestimation of recession curves (e.g, the Kushum and Gram stations in Figure 13)

Both the cases mentioned above were taken into account during the extreme values analysis and adjustment as described in Section 4.

For the several small ungauged catchments, we applied the soil and surface parameterization calibrated in the gauged catchments. The extension of the parameterization from gauged to ungauged catchments was based on the soil type and land use classification.

Figure 13 shows observed (blue line) and TOPKAPI simulated (red line) discharge for a series of relevant catchments and locations, along with temperature and precipitation records. As expected, the presence of reservoirs and lakes has a clear impact on the simulation performance: the sites of Turkmenabat and Atamurat on the Amu Darya River for example, are located downstream a series of reservoirs, which are not modeled in TOPKAPI. This causes the model to overestimate

discharge in spring/summer when the reservoirs retain part of the snowmelt, and underestimate discharge in the rest of the year, when the reservoirs slowly get emptied as shown in Figure 13. A similar behavior is observed for the sites of Semiyarskoje and Chirchik on the Irtysh and the Syr Darya rivers, respectively. The two stations are both located downstream of a lake (Zajsan lake's area is 1.810 km^2, Charvak lake's is 40 km^2), a fact that explains the shift and delay in streamflow not observed in the respective catchment's upstream sites. The stations at Orenburg (Ural river) and Ulba Perevalochnaya (Amu Darya river) represent optimal situations where the model is able to reproduce well the discharge over the entire period. The station at Kushum (Ural river) is located within an alluvial plain, whose lamination effect cannot be reproduced by the TOPKAPI model and we allowed the model to overestimate the peaks and underestimate the recession curves.

Figure 14 shows a comparison of observed and simulated maximum flows for a series of selected sites. We observe how TOPKAPI simulated annual maximum discharge shows a good correlation and a low bias for most of the locations.

The known limitations of the hydrological modeling over this large area (namely effect of lakes/reservoirs and alluvial plains) will be eventually corrected to better match the available observations during the extreme value analysis (Section 4) and to finally derive realistic input hydrographs for the hydraulic modeling component (Section 5).

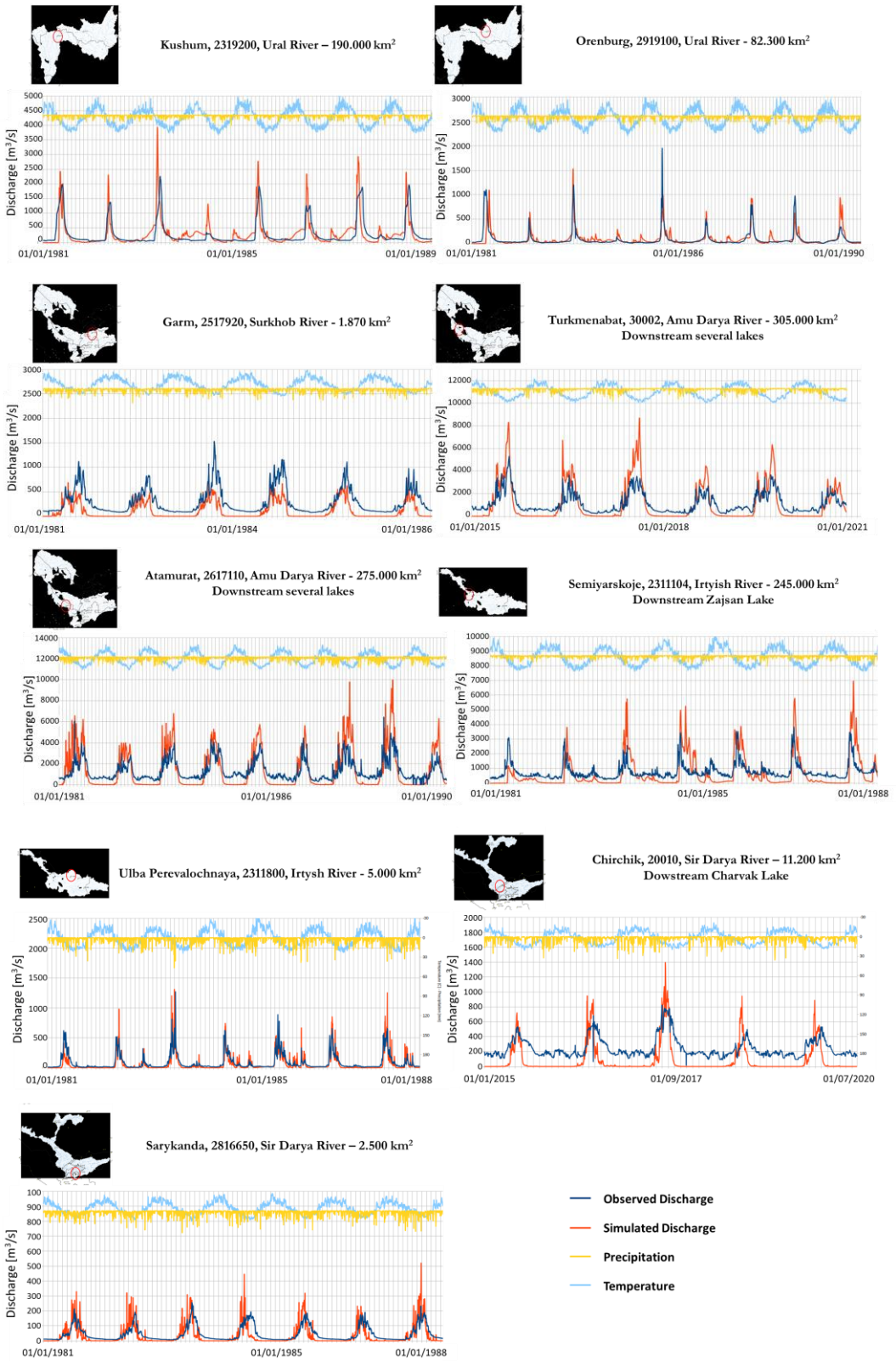


Figure 13. Example of observed (blue line) versus TOPKAPI simulated (red line) discharge computed at various locations for selected river sections. The effect of dams (e.g., Turkmenabat and Atamurat) and floodplains (e.g., Kushum) can be observed in some locations, but the flow peaks are overall well reproduced.

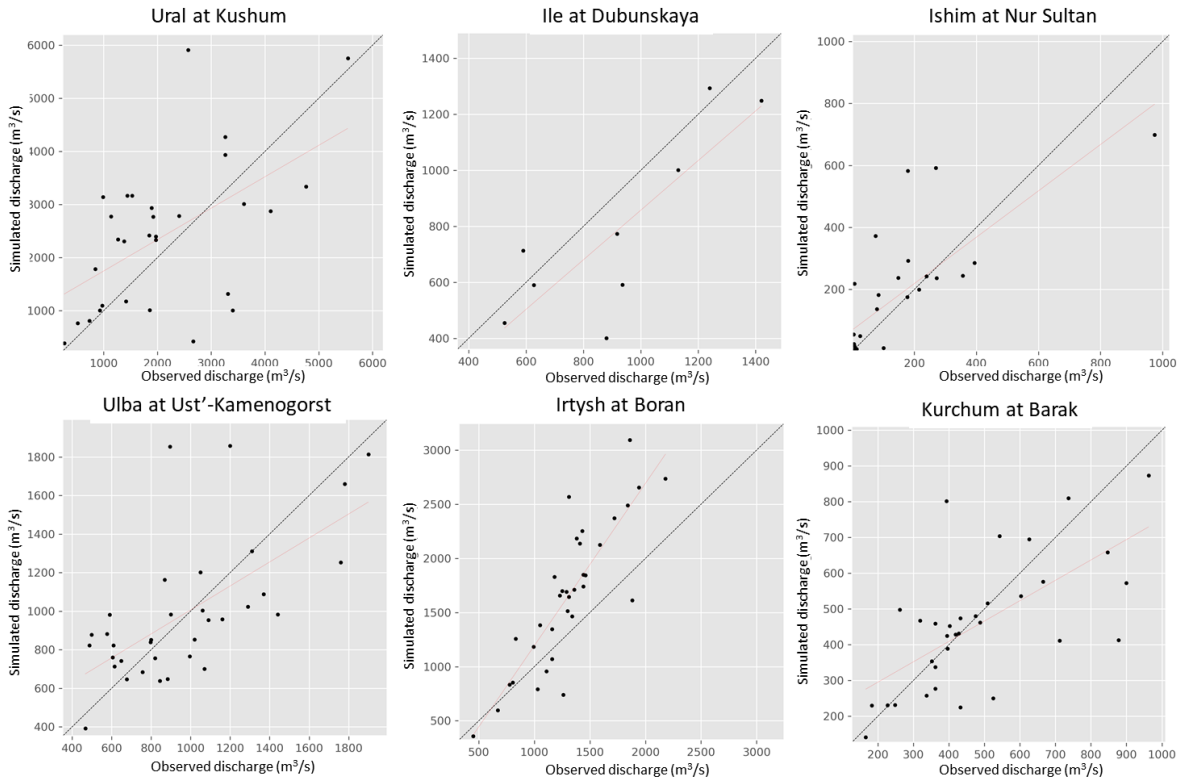


Figure 14. Observed versus TOPKAPI simulated maximum annual discharge for selected sites. The black lines indicate the 1:1 line, the red lines indicate the trendline fitted on the scatterplot.

Trend lines in Figure 14 show that the simulated annual maxima for some stations are in some cases biased compared to the observed maxima. This issue has been dealt with in the extreme value analysis through an adjustment of the simulated annual maxima, and is described in the next section. Percent bias values between observed and simulated annual maxima are provided in the next section, in Table 5.

Table 4 shows the correlation coefficients computed between observed and simulated annual maximum flows for stations with more than 15 years of data (post 1980) and more than 10,000 km^2 drainage area. The median correlation is 0.46.

Table 4. Correlation coefficients between observed and simulated annual maximum flows for observed stations with more than 15 years of data and more than 10,000 km^2 drainage area.

Name	Basin	Longitude	Latitude	Drainage area	Number of years with records (post 1980)	Correlation
Below Ravotkhodzha dam	Zeravshan	67.4	39.53333	12300	15	0.71
Stone quarry	Ishim	66.267	52.02	86200	22	0.12
West Toksanby	Ishim	66.628	52.963	90000	23	0.34
Sergeevka	Ishim	67.432	53.879	109000	34	0.62
Peter and Paul	Ishim	69.125	54.915	118000	33	0.72
Dolmatova	Ishim	69.445	55.317	142000	24	0.72
Kalkutan-Kalkutan	Ishim	69.464	51.786	16500	33	0.30
Boran	Ishim	85.222	48.005	55900	34	0.85
Ust-Kamenogorsk	Irtysh	82.716	49.902	146000	34	0.39
Semyyarka	Irtysh	78.312	50.886	229000	32	0.56
Pavlodar	Irtysh	76.883	52.329	240000	18	0.33
Forest	Irtysh	84.329	49.83	10700	34	0.72
Nura - Sheshenkara village	Nura	74.053	49.969	13980	34	0.43
Nura - Koshkarbayev village	Nura	71.336	50.829	50760	34	0.73
Will	Emba	54.672	49.089	17100	37	0.62
Saga	Emba	57.43	48.487	16100	34	0.16
Ushtobe	Karatal	77.967	45.221	13200	16	0.68
Kushum	Ural	51.143	51.865	190000	30	0.55
Atyrau	Ural	51.878	47.089	236000	31	0.60
Chilik	Ural	54.078	51.119	37300	18	0.70
Ushzharma	Ile	75.803	44.957	129000	15	0.05
Gazalkent dam	Syr Darya	69.75	41.56667	112000	15	0.75
Shardara Reservoir	Syr Darya	67.954	41.265	174000	34	0.50
Tomenaryk	Syr Darya	67.023	44.017	219000	34	0.28
Arys	Syr Darya	68.849	42.445	13100	34	0.39
Naryn,73-Naryn City	Syr Darya	76.01991	41.42397	10500	37	0.19
Naryn,-mouth of Kekdzherty river	Syr Darya	75.67852	41.42281	14900	15	0.38
Uchkurgan	Syr Darya	72.1	41.15	58400	15	0.07

4 Extreme Value Analysis and Adjustment

In order to assess fluvial flood hazard, we need to characterize the statistical behavior of extreme flows through the analysis of annual maxima. This is done using the simulated discharge data produced during the hydrologic simulations together with historical discharge data collected from local and global sources (see Table 2 and Table 3). A great wealth of flow data is available to be exploited, especially in high-and medium-exposure areas, with some very long historical records (see Figure 3 and Figure 4) even dating back to the 1930s-40s-50s.

The hydrological model was used to obtain a spatially distributed representation of the fluvial flood hazard that covers homogeneously all the geographical extension of the target countries. This goal was impossible to attain using only observed data. However, the model results have some limitations, summarized here, and described in detail in the next paragraphs:

- Spatial changes in extreme value behaviors: rivers can significantly alter the patterns of extreme flows as the flow downward, especially if they are crossing a large alluvial plain, with low to almost null slopes and space for inundation. This is often the case in Central Asia, where very large alluvial plains crossed by major rivers exist. Hydrological models are generally not suited to reproduce accurately the peak flow reduction caused by alluvial plains (see Section 4.1.2).
- Temporal changes in extreme value behaviors: the construction of large reservoirs typically reduces peak flows and alters temporal extreme flow patterns, but accurate data for the characterization of reservoirs (depth vs volume curves, spillways and outlet shape, operational rules, etc.) are rarely available (see Section 4.1.2).
- Model uncertainty: despite having been calibrated against observed data, the results of the hydrological model are unavoidably affected by errors and biases (as is always the case with mathematical models) (see Section 4.1.3).

To overcome these limitations, historical data can be exploited. Historical discharge data reveal temporal shifts in flow that are observed after a man-made hydraulic structure is built (e.g., dam, regulated reservoir). Historical discharge data are also useful for accurately representing flow behavior in Central Asia large floodplains. Observing the spatial shifts in the extreme flow values signal, it is possible to represent the natural peak flow reduction consequent to the flood water expanding in the floodplains, a phenomenon that cannot be simulated through hydrological modeling. Historical data were also used to identify potential modeling errors and correct them.

4.1 Methodology

4.1.1 GEV fitting

The generalized extreme value (GEV) distribution, introduced by Jenkinson in 1955 (Jenkinson, 1955), is a standard tool for modeling flood peaks using annual maximum series (Morrison and Smith, 2002; Rosbjerg and Madsen, 1995). The GEV distribution combines into a single form all three Extreme Value (EV) distributions: the Gumbel, the Frechet and the Weibull. The GEV cumulative distribution has the following form:

$$F(x,\mu,\sigma,\xi) = exp\left(-\left(1+\xi\left(\frac{x-\mu}{\sigma}\right)\right)^{-1/\xi}\right)$$

Where the three parameters called shape (ξ), location (μ) and scale (σ) are estimated from data. Parameter estimation procedures for the GEV distribution have been extensively studied (Martins and Stedinger, 2000). The most commonly used methods are the maximum likelihood estimation (MLE) (Prescott and Walden, 1980), non-linear least squares (Teunissen and Knickmeyer, 1988), the method of L moments (LMOM) (Hosking, 1990), and the method of moments (MOM) introduced by Karl Pearson in 1894.

We used both observed and simulated annual maximum discharge values to fit the GEV using the non-linear least squares method.

4.1.2 Temporal and spatial shifts detection

When hydraulic structures are built to regulate the natural streamflow (e.g., for flood protection, agriculture, hydropower production) a temporal shift is observed in the discharge records corresponding to the time of construction. Figure 15 shows the temporal shift in discharge on the Syr Darya at Tomenaryk relative to the construction of the Shardara Dam which was built 400 km upstream in 1967.

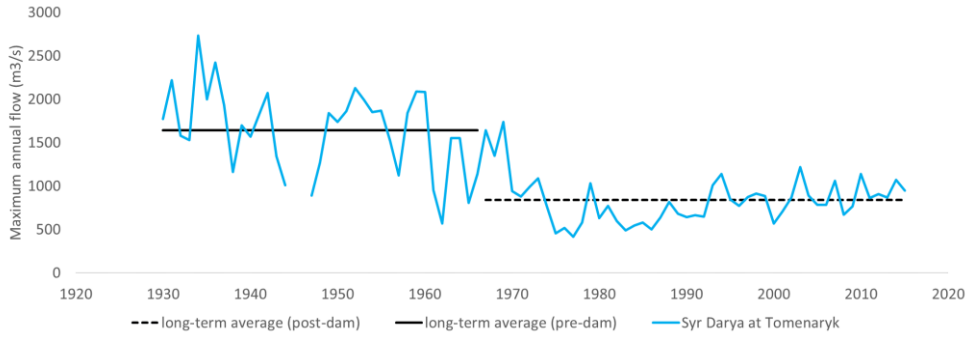


Figure 15. Example of temporal change in the hydrological regime observed on the Syr Darya at Tomenaryk caused by the construction of the Shardara Dam in 1967

In order to accurately represent flood hazard, the new flow regime should be represented. Since it was not possible to model the regulated regimes mechanistically due to lack of reservoirs operation data, we used discharge data to detect these changes and we applied them to our simulated flow data.

Temporal shifts produced by dam constructions were first identified by locating all the dams in the area through the Global Reservoir and Dam Database (GRanD) (The Global Water System Project, 2011) and the other local sources of information, and then by analyzing the daily discharge observed data downstream the dam before and after the construction. The GRanD provides a set of relevant information on regulated lakes and reservoirs including their location, first year of operation and downstream affected flow gauges (Figure 16). This database was integrated with other sources, including specific websites and satellite data. With an analysis of the discharge data from the downstream stations around the date of dam construction, it is possible to identify the temporal shift and apply it to our simulated streamflow values.



Figure 16. Dams' location in Central Asia from the GRanD database and other sources of information.

Similarly, we adjusted our simulated flow data to introduce the effect of floodplains on peak flow reduction. A spatial shift in extreme flow values signal is naturally observed in consecutive stations when the water expands in large floodplains. Figure 17 shows the detection of peak flow reduction in consecutive gauging stations on the Ishim River while it creates a floodplain between Sergeyevka lake and the border with Russia.

The methodology used for such adjustment is described in the next section.

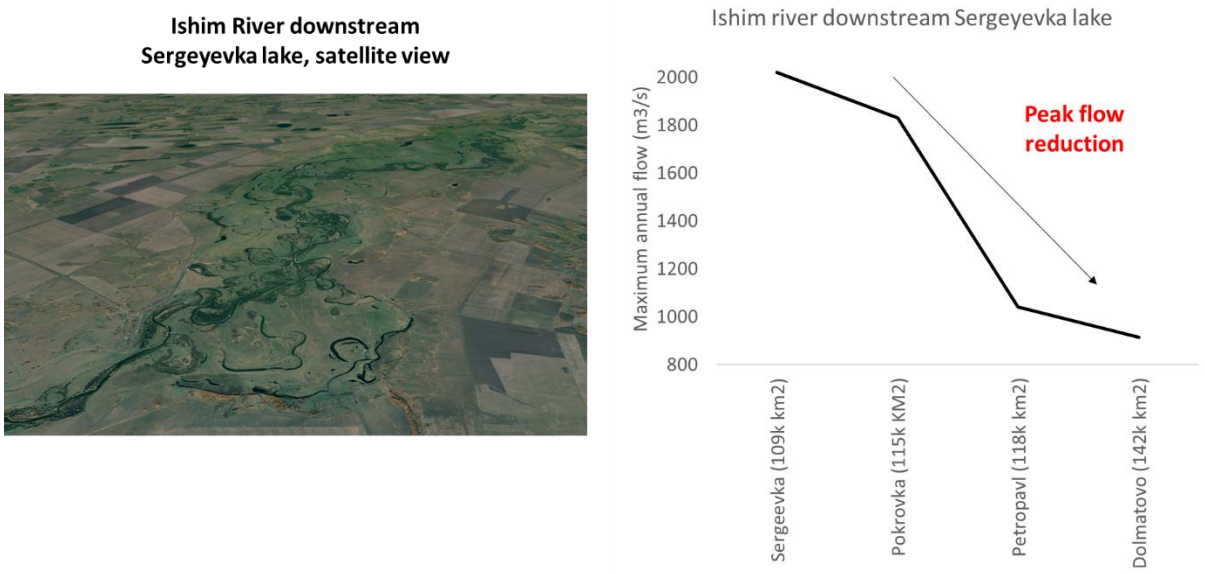


Figure 17. Example of spatial shift in maximum annual flow signal observed in consecutive gauging stations on the Ishim River, downstream the Sergeyevka lake

4.1.3 Extreme values adjustment and extrapolation

Adjustment of simulated flows was based on the comparison of observed and simulated extreme values distributions at gauged points. We observed that the largest hydrological model errors occur on the main stem of large rivers, because of the impact of large reservoirs and floodplains.

Model bias and temporal/spatial shifts can be identified and corrected when long records of discharge data are available, as described in the previous sections. For the stations where observed data were not available, we extrapolated the extreme values adjustment from the available stations based on their location and flow accumulated area.

Where observed data were available, the adjustment was done by computing the ratio between observed and simulated mean annual maximum flows and then multiplying the simulated flow for such ratio. In other terms, the simulated annual maximum flows were increased or decreased by a coefficient based on the mean bias between observed and simulated mean annual maximum flows.

Where observed data were not available, the adjustment coefficient was computed by using an adjustment coefficient from an associated station, where the association was carried out based on their location and flow accumulated area.

Figure 18 shows the overall goodness-of-fit values (Kolmogorov-Smirnov indicator or K-S (Massey, 1951)) for both the uncorrected (blue) and corrected (orange) peak flows when compared to the observed values. The K-S test is a nonparametric test that is commonly used to compare two samples and it quantifies the distance between the quantiles of the empirical distributions of

those two samples. In this sense, a low value indicates a good fit. The observed median values of our K-S test are 0.59 for the uncorrected and 0.24 for the corrected peak flow values.

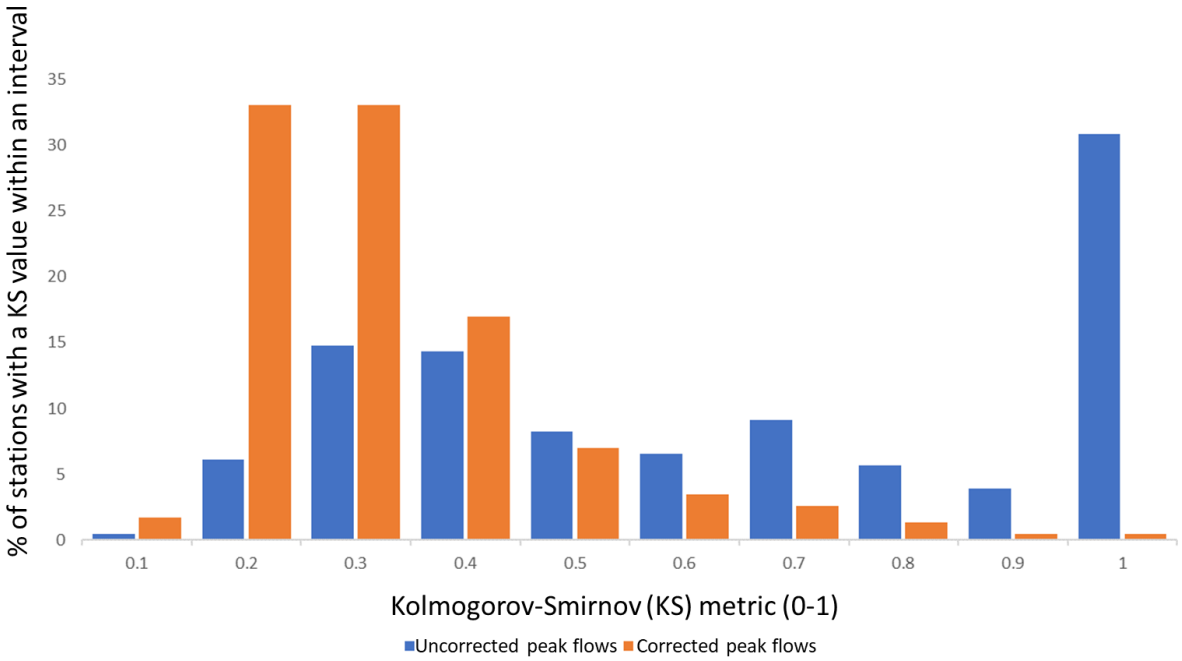


Figure 18. Overall goodness-of-fit (Kolmogorov–Smirnov indicator) for uncorrected (blue) and corrected (orange) peak flows.

Table 5 shows the percent bias between observed and simulated annual maximum flows for the stations in Table 4, before and after the adjustment. The pre-adjustment bias shows a systematic positive bias, which was expected based on the hydrological model calibration strategy we adopted (see Section 3.2), allowing the model to overestimate the peaks in alluvial floodplains and downstream reservoirs. The systematic positive bias is eventually removed after the adjustment. It is worth mentioning that the correlations between observed and simulated annual maximum flows showed in Table 4 remains unchanged, since the adjustment is applied on the mean of annual maximum flows.

Table 5. Percent bias (before and after the adjustment) between observed and simulated annual maximum flows for observed stations with more than 15 years of data and more than 10,000 km² drainage area.

Name	Catchment	% bias (pre-adjustment) [%]	% bias (post-adjustment) [%]
Below Ravotkhodzha dam	Zeravshan	61	-11
Stone quarry	Ishim	22	16
West Toksanby	Ishim	-24	-19
Sergeevka	Ishim	17	-3
Peter and Paul	Ishim	108	-9
Dolmatova	Ishim	164	-2
Kalkutan-Kalkutan	Ishim	23	-14
Boran	Ishim	27	11
Ust-Kamenogorsk	Irtysh	214	55
Semyyarka	Irtysh	45	7
Pavlodar	Irtysh	91	18
Forest	Irtysh	0	-1
Nura - Sheshenkara village	Nura	77	-8
Nura - Koshkarbayev village	Nura	-1	-3
Will	Emba	50	51
Saga	Emba	17	16
Ushtobe	Karatal	-6	-2
Kushum	Ural	14	17
Atyrau	Ural	100	9
Chilik	Ural	-68	56
Ushzharma	Ile	53	2
Gazalkent dam	Syr Darya	63	-9
Shardara Reservoir	Syr Darya	290	3
Tomenaryk	Syr Darya	425	7
Arys	Syr Darya	84	30
Naryn,73-Naryn City	Syr Darya	-35	6
Naryn,-mouth of Kekdzherty river	Syr Darya	-37	1
Uchkurgan	Syr Darya	41	-11

Figure 20 shows some examples of bias-correction on the hazard curves computed at four relevant locations.

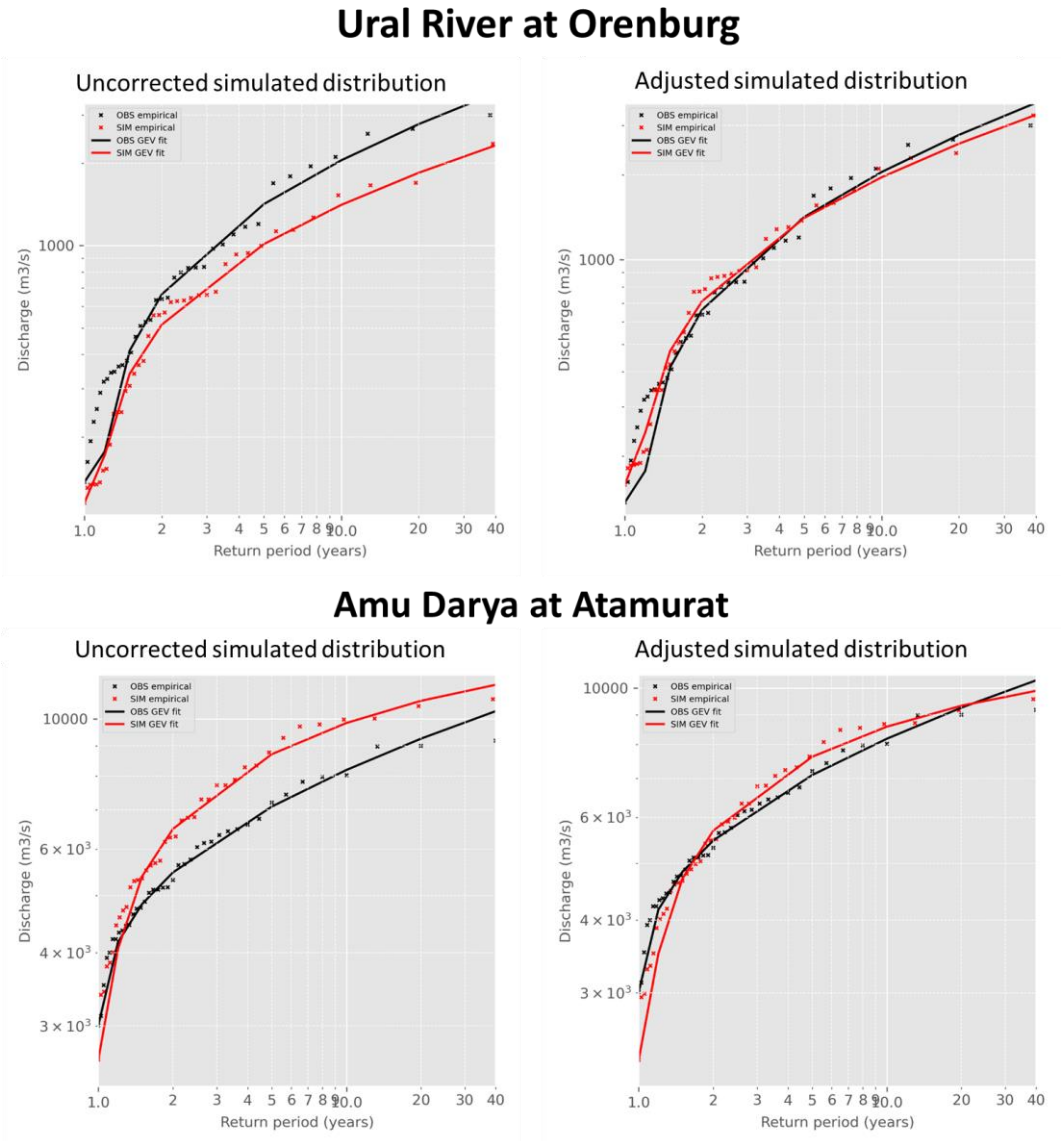


Figure 19. Example of adjustment of discharge extreme value distributions at selected sites

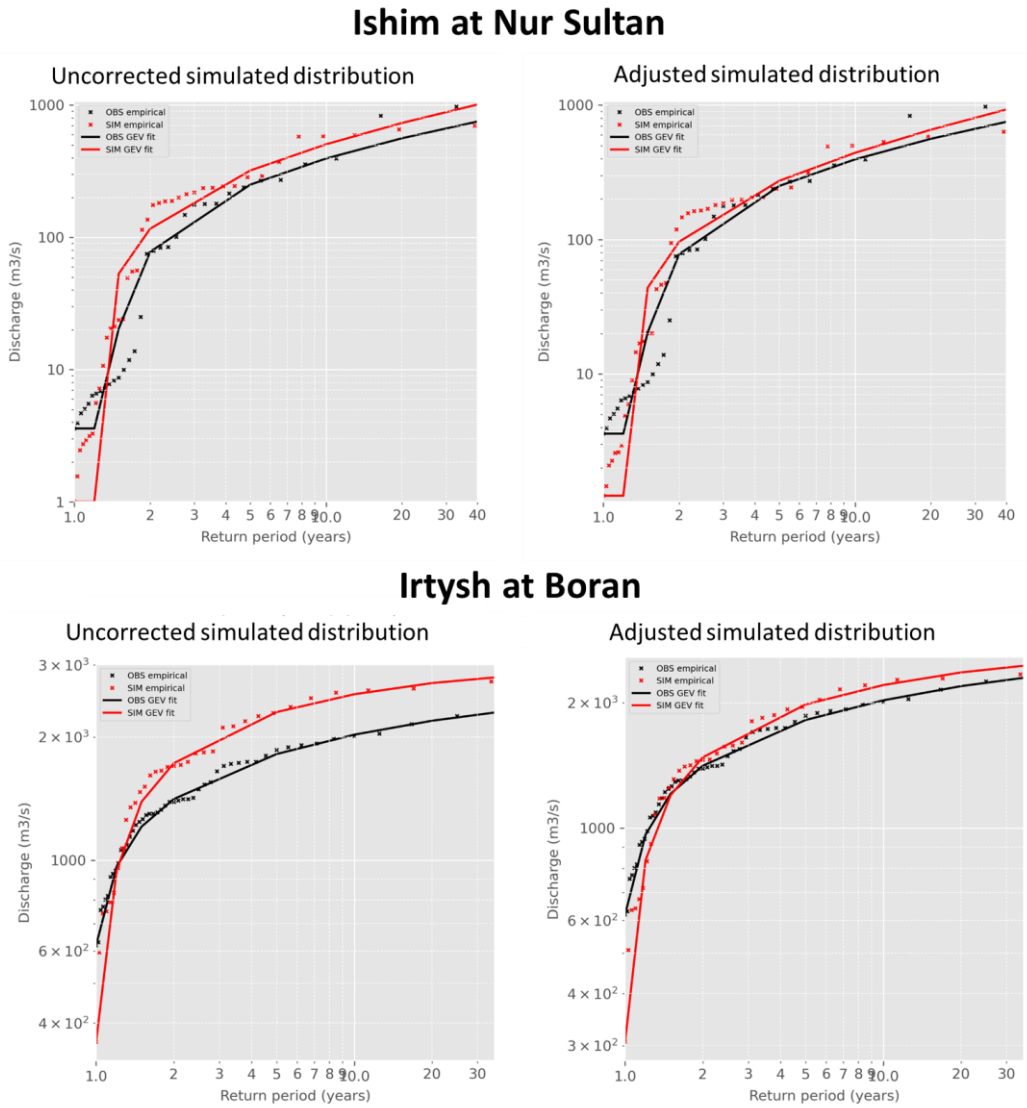


Figure 20. Example of adjustment of discharge extreme value distributions at selected sites

4.2 Output

The final result of the Extreme Value Analysis was the extreme flow value distribution for all the virtual stations (i.e., the simulation points), from which peak flows for the flood simulation are extracted (1-in-5, 10, 20, 50, 100, 200, 500, 1000 years floods) (see Figure 21). In total, we identified 78,000 virtual stations over the entire region. These values are used as peak flow values to build input hydrographs for the CA2D hydraulic model.

station_id	X	Y	2	5	10	20	50	100	200	500	1000	2000	5000
2617110	-154680	-905090	1778	2457	2971	3486	4165	4679	5193	5873	6387	6901	7581
2217100	116320	-999090	244	337	407	478	571	642	712	805	876	946	1039
2217101	163320	-1030090	155	214	258	303	362	407	452	511	555	600	659
2217102	150320	-1096090	128	177	214	251	300	337	374	423	460	497	545
2217103	145320	-1153090	81	113	136	160	191	214	238	269	293	316	347
2217104	83320	-1185090	1	1	1	1	1	2	2	2	2	2	3
2217110	89320	-1192090	32	45	54	64	76	85	95	107	116	126	138
2217120	154320	-1151090	24	34	41	48	57	64	71	80	88	95	104
2217201	344320	-998090	50	69	84	98	117	132	146	165	180	194	213
2217130	197320	-1026090	65	90	108	127	152	171	189	214	233	252	276
2517550	158320	-802090	8	11	13	15	18	21	23	26	28	31	34
2517920	288320	-771090	127	175	212	248	297	333	370	418	455	491	540
2817100	-598680	-382090	5596	6596	6596	6596	6596	6596	7046	7968	8666	9363	10285

Figure 21. Example of peak flow values at different return times (5, 10, 20, 50, 100, 200, 500 and 1000 years) for a few virtual stations identified by their ID

It is worth mentioning that the results of the hydrological model and the adjustment procedure are affected by errors. The main sources of errors include the representation of the precipitation and temperature, the errors in the observed flows used for calibration, the limitation of the mathematical models used in this project, which introduce necessary simplification to the natural processes, and the natural variability of the rainfall-runoff mechanism. Such errors translate into uncertainty of the results. Given the scale of the modeling application, which covers five large countries, it is safe to assume that in some cases such uncertainty will be significant and must be quantified in order to produce results that can be used correctly in guiding and assisting disaster risk reduction policies. This aspect is discussed and quantified in Section 5.4 of this report.

5 Hydraulic Modeling and fluvial flood hazard maps derivation

The CA2D hydraulic model was developed and validated by Dottori and Todini (Dottori and Todini, 2011) and uses a 2D Cellular Automata (CA) approach. A Cellular Automata (CA) is a cell-centered, finite volume scheme for the description of physical dynamic systems regulated only by local laws. In this scheme, each cell assumes a certain state (or property), which evolves in time according to specified fixed rules. In CA2D case, the momentum equation is solved for each time step, computing volume exchanges between grid cells along the cell's borders. Volumes of each cell are successively updated using flow volume conservative equations. We ran the model using the semi-inertial formulation of the momentum equation, which was developed for the LISFLOOD-FP model (Bates et al., 2010). This approach allows for high resolution simulations at a significantly reduced computational effort, making it possible to run hydraulic simulations at the continental and global scale (Dottori and Todini, 2011).

5.1 Model Setup

The consortium holds an internal version of the CA2D model, which was implemented in C++ programming language and fully parallelized via Graphics Processing Units (GPUs) to improve its performance in terms of computational time. The model was run with a 0.00083° (~90 m) spatial resolution. The simulation time-step is dynamic, and it varies between 0.01 and 15 seconds, with the maximum allowable time step defined by the Courant–Friedrichs–Lewy or CFL condition, which is commonly adopted to preserve stability in computational fluid dynamics models.

The stochastic catalogue of streamflow produced via the Extreme Value Analysis (see Section 4) was used to derive the simulated flood hydrographs that serve as input to CA2D. The hydraulic model also takes spatially distributed data as inputs, such as the DEM and roughness to define the computation grid and determine the flow diffusion and velocity. For this purpose we used the MERIT DEM (Karnieli et al., 2010) (Figure 5) together with MERIT Hydro (Yamazaki et al., 2019). Land use maps for GlobeLand30 (Figure 7) were used instead to derive roughness values.

Flow diffusion describes the water volume exchanges between grid cells, by integrating the semi-inertial momentum equation along the cells' connections which are called links. The computation grid can have various link network configurations, in our application, we adopted a four direction link network, which is better suited for large scale studies and Central Asia's geomorphology (Dottori, 2012). Then, flow volumes are updated at each time step following the conservative equation. The result is a flood footprint of the maximum water depth at each cell over the simulation.

Figure 22 schematically represents the structure of the CA2D hydraulic model.

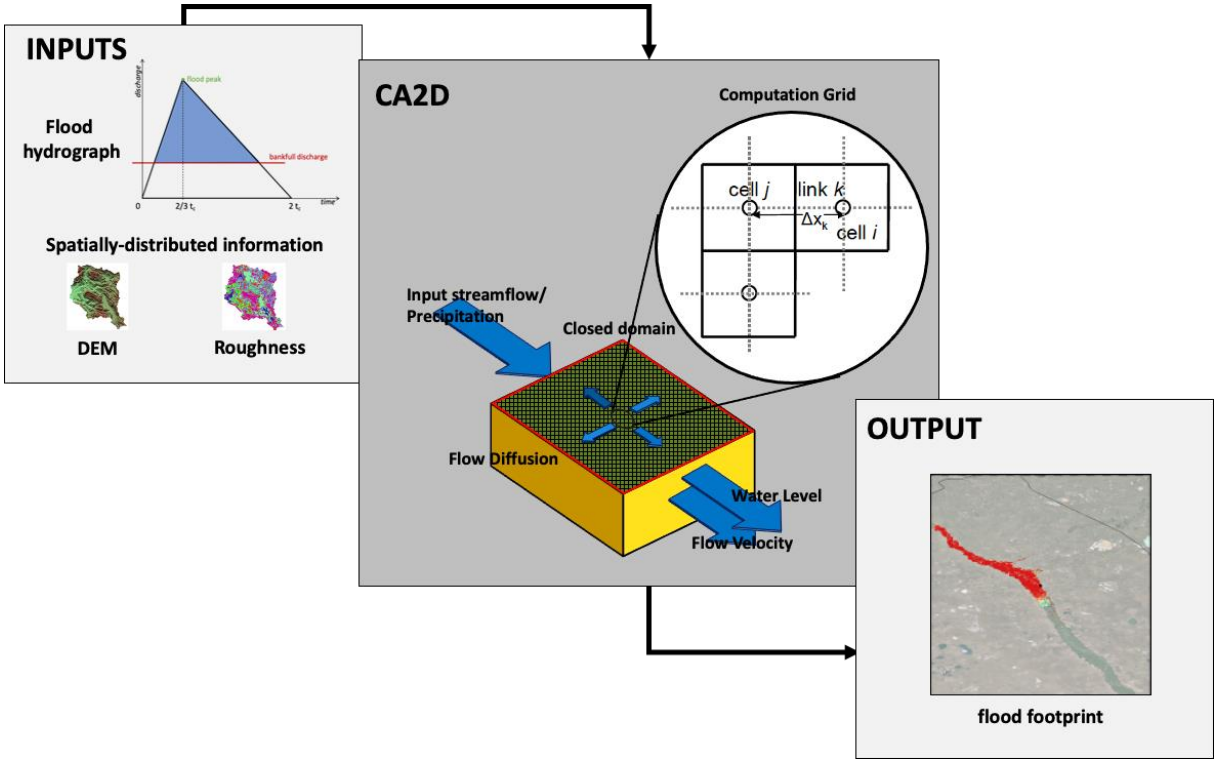


Figure 22. Conceptual scheme of the CA2D hydraulic model

5.2 Model configuration

5.2.1 Spatial discretization

In order to run the model, we first discretize the simulation domain into units. We run a simulation for each spatial domain unit in order to take full advantage of the model parallelization. A simulation unit is a squared tile, which includes the hydraulic inlet from which the water flows into the domain. Tiles have open boundaries with critical flow condition, which means that when water reaches the tile boundary, it flows out of the simulation domain.

We define the virtual stations that will serve as inlet in the hydraulic simulation with the following criteria:

- they have flow accumulated area greater than 100km^2
- their distance varies between 5 and 40 km depending on their flow accumulated area

Each virtual station is characterized by maximum discharge values at various return periods, identified in the Extreme Value Analysis (see Section 4).

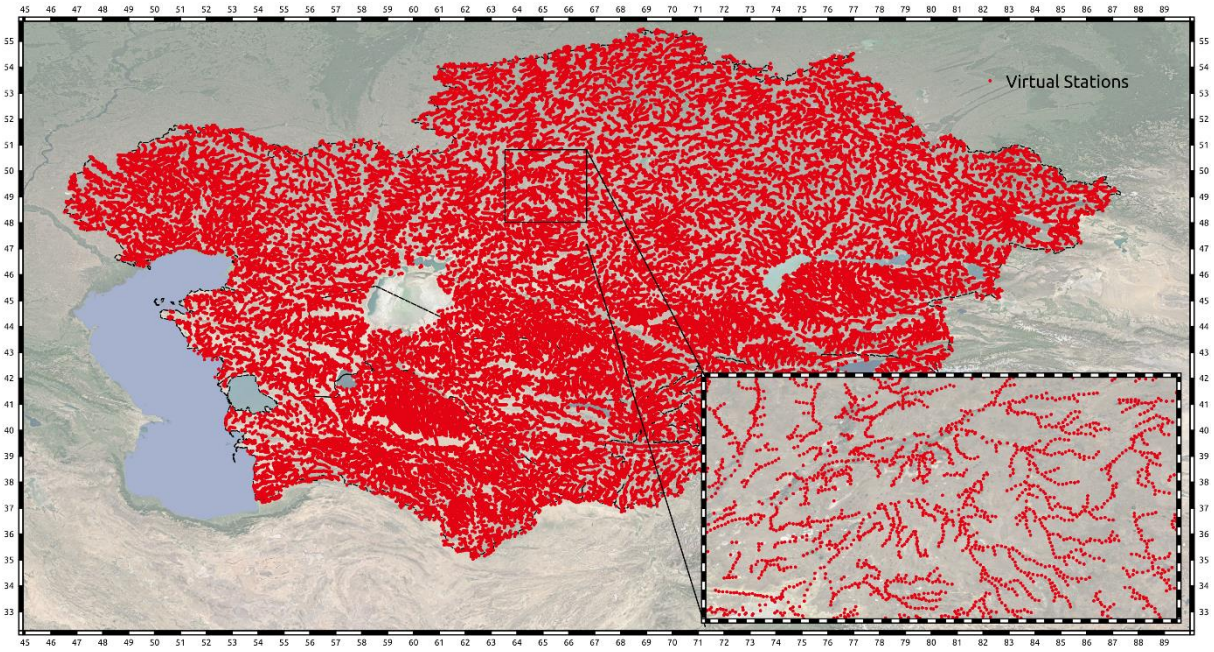


Figure 23. Distribution of virtual stations over the entire region

According to the previous criteria around 78,000 virtual stations are identified over the entire region of Central Asia (Figure 23). Each of these virtual stations is then associated to a tile that represents the simulation domain for the hydraulic model CA2D (see Figure 24). Having a series of small tiles allows us to parallelize the simulation using GPUs and, therefore, to significantly reduce computational time. The squared tiles overlap and together cover the entire area. The overlapping is functional to obtaining homogenous results, as when the final hazard map is derived, the maximum water depth value is taken for each grid cell.

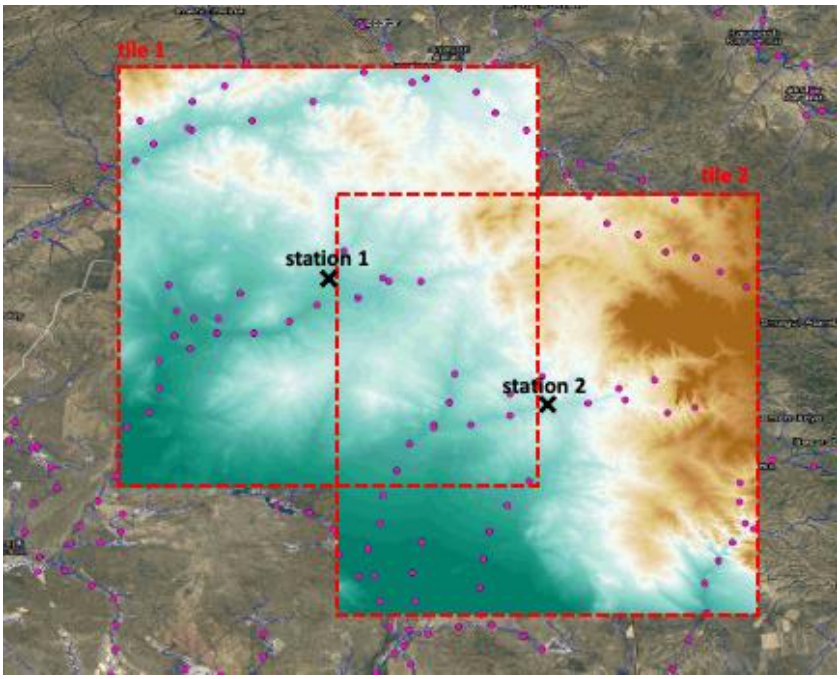


Figure 24. Schematic representation of CA2D simulation domain units

Each tile is characterized by:

- map of elevation (from MERIT DEM)
- map of roughness (derived from GlobeLand30 land use maps)
- inlet hydrograph in the correspondent virtual station location and derived through the Extreme Value Analysis (see Section 5.2.3)

5.2.2 Deltas

We use a different strategy for deltas. In deltas water flows from the main river channel to the various distributary channels but TOPKAPI does not reproduce this behavior. Instead, the hydrologic model routes the entire flow along the river network, up to the river mouth. To model the main deltas properly, we identified a series of custom polygonal tiles, covering the entire delta fan (see Figure 25). Only a few major inlets are simulated (e.g., the ones in the red dots in Figure 25), and the water is distributed in the delta during the hydraulic simulation. This strategy has been used to derive the flood footprints over the Ural, Emba, Amu Darya, Sir Darya, Zeravshan and Ili rivers.

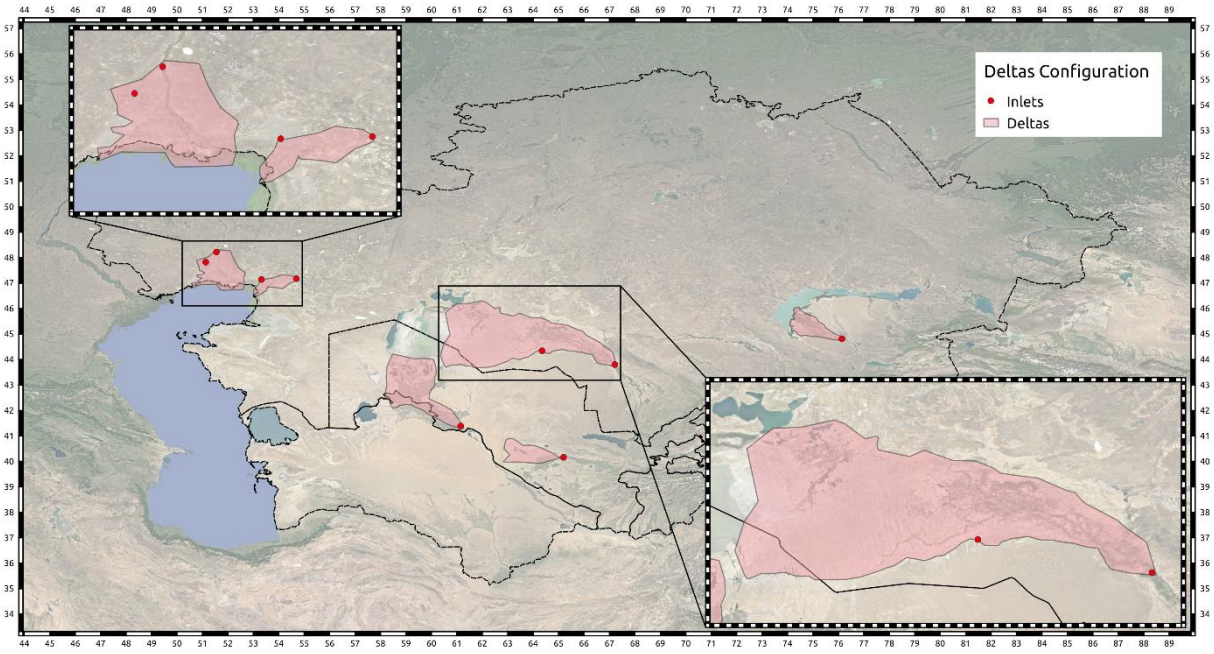


Figure 25. Configurations of deltas custom tiles with zoom over Ural, Emba and Sir Darya rivers

5.2.3 Flood hydrographs

For each tile, we built a flood hydrograph based on the virtual station's maximum discharge at various return periods calculated during the extreme value analysis. We assume the bankfull discharge as the discharge at 2-year return period and simulate the maximum discharge at years 5, 10, 20, 50, 100, 200, 500 and 1000.

To build the hydrograph, we calculate the time of concentration (t_c) defined as the time needed for water to flow from the most remote point in a watershed to the watershed outlet. The value of t_c

is a function of the watershed topography and is calculated via the Giandotti formula (Giandotti, 1934):

$$t_c = \frac{4\sqrt{S} + 1.5L}{0.8\sqrt{H_m - h_0}}$$

Where S is the accumulated area, L is the length of the river, while H_m and h_0 are the average catchment altitude and the altitude at the virtual station location, respectively.

As illustrated in Figure 26, we assume a triangular hydrograph reaching the flood peak at 2/3 t_c and going back to zero at 2 t_c. From this hydrograph, we remove the bankfull discharge that is assumed to remain in the channel.

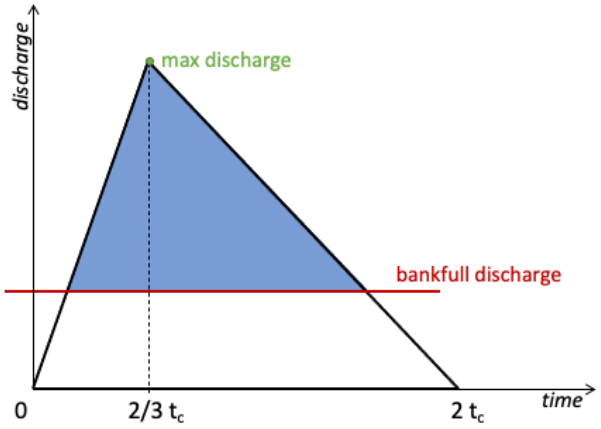


Figure 26. Definition of flood hydrograph provided as input to the CA2D hydraulic model

5.2.3.1 Validation

To validate our methodology, we extracted for a series of historical flood events the flood peak hydrographs from the available daily discharge records and compared them with the triangular hydrographs built as per our assumption.

Figure 27 shows some examples of triangular hydrographs (orange line) and their relative historical hydrographs (blue line) at selected stations in the Ural, the Irtysh, and the Murghāb River. Despite the specificity of each hydrograph due to the real temporal occurrence of the precipitation that generated it, our methodology seems to perform well both in terms of flood peak dynamic and water volume. The performance is consistent for medium-to long t_c values (Ural and Irtysh) as well as for shorter ones (Murghāb). We also observed consistency in time, with a good hydrograph shape reproduction for two proximal stations in the Irtysh River for records collected in different decades (second panel Figure 27).

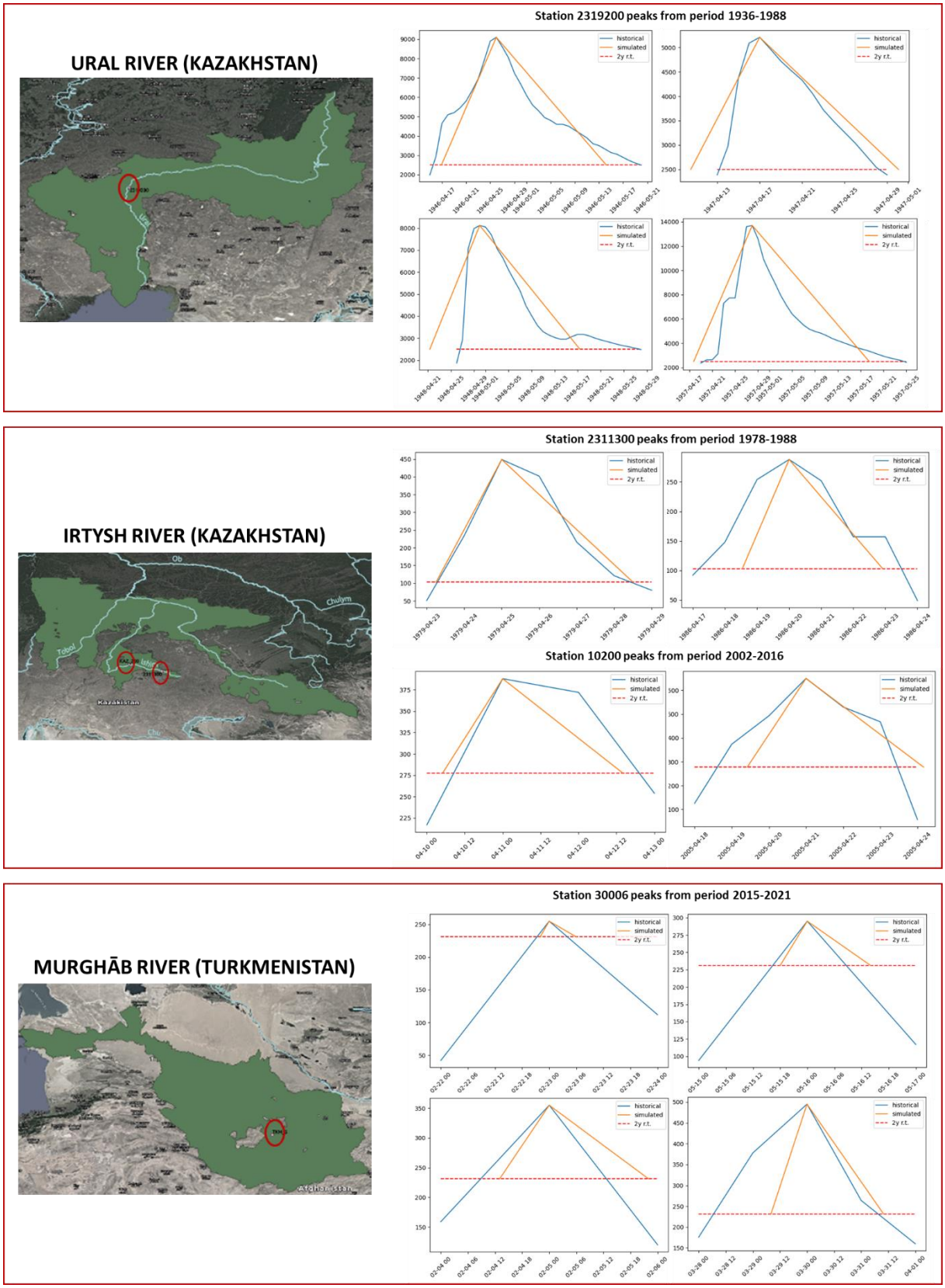


Figure 27. Examples of triangular hydrographs (orange) and corresponding historical hydrographs (blue) at selected stations for a set of selected flood events

5.2.4 Derivation of the hazard maps

In order to derive the hazard maps over the entire domain, for each return period a CA2D simulation is performed for each tile, feeding the model with the above-described hydrographs. Figure 28 shows an example of one tile used as domain for the CA2D simulation of flood footprint caused by an event with 100-year return period for a generic virtual station (black circle).

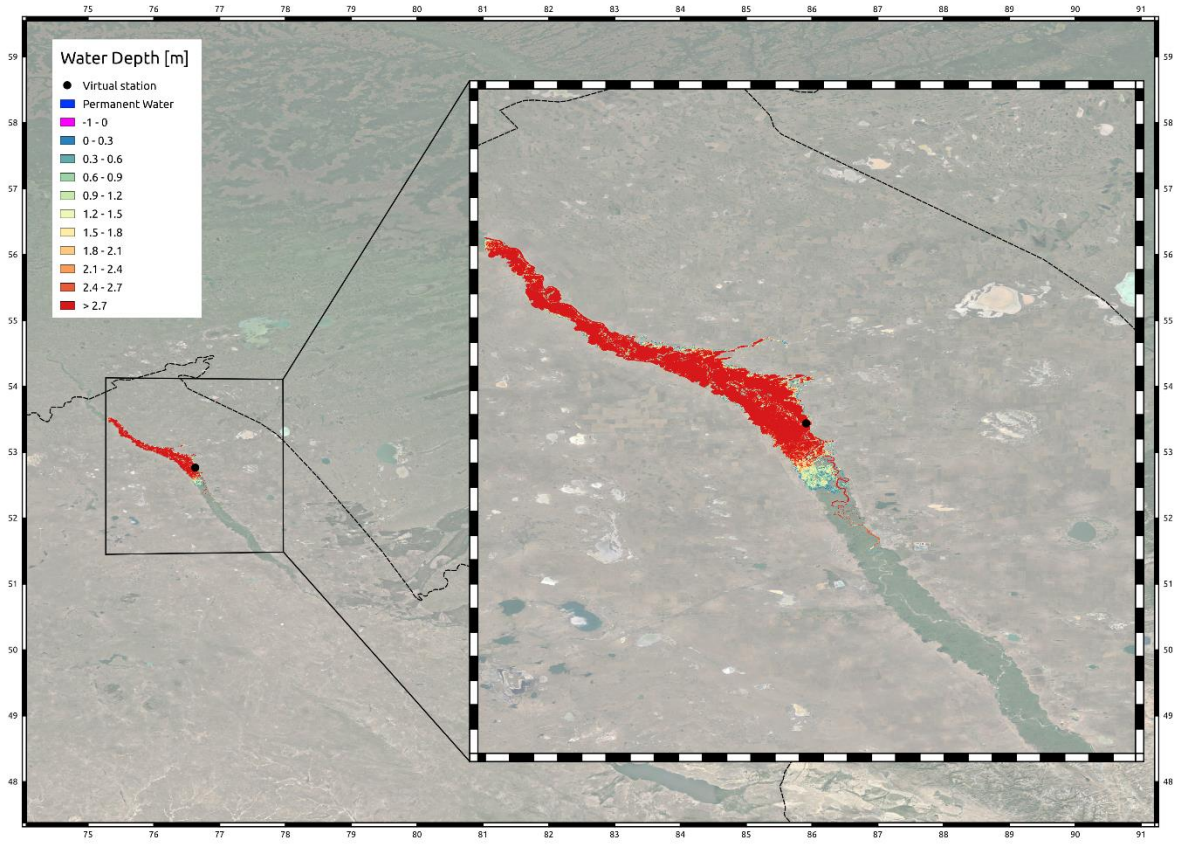


Figure 28. Example of a tile associated to a virtual station. The area in the inner panel represents the tile and the CA2D domain simulation for the corresponding virtual station (black dot), water is flowing to NW.

To obtain the hazard map for a specific return period, the outputs from the CA2D simulations of all tiles obtained with an input hydrograph associated to the discharge with the same return period are eventually merged by taking the maximum water depth value for each grid cell. The resulting hazard maps are presented for the return period of 10, 100 and 500 years in Annex 1.

Figure 29 shows the derived hazard curves for five selected target sites across the region. In particular the locations have been selected within the urban areas of the main flood-prone cities of the target countries: Turkmenabat, Tashkent, Dushanbe, Nur Sultan, Bishkek.

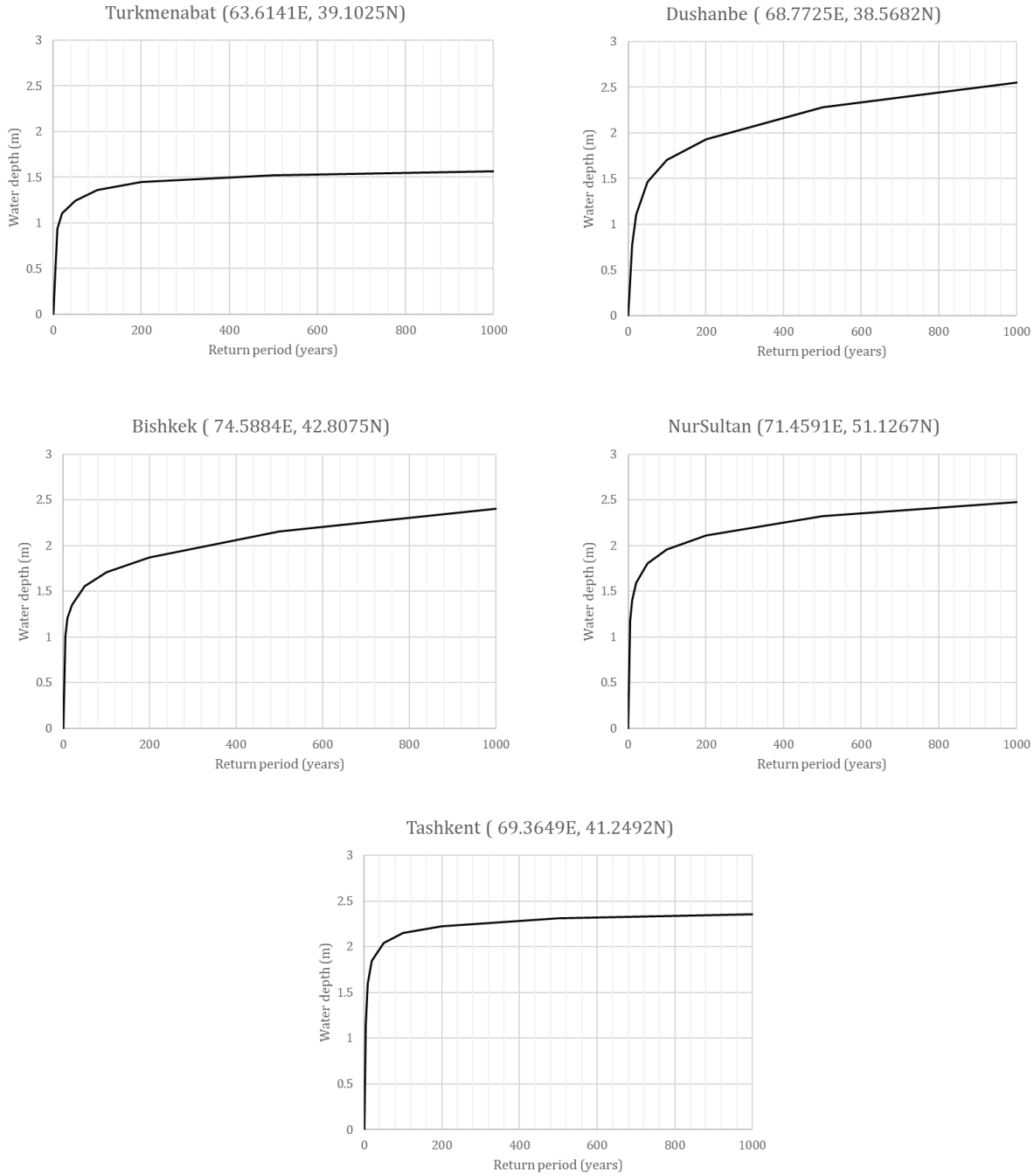


Figure 29. Undefended flood hazard curves computed at five selected target sites

5.3 Validation

Since water depth data observations are not available, it is not possible to numerically validate the output of the hydraulic model. However, the CA2D model has been extensively validated in other applications (Dottori and Todini, 2011) and all the previous modelling steps have been validated for Central Asia.

Alternatively, validation of the hydraulic model approach is carried out through the comparison with available historical flood events, which are selected considering the availability of the following information:

- Daily discharge records covering the flood event
- Flood footprint map, either derived from satellite products or from direct accounts.
- Damage reconnaissance reports

We have searched data for a series of potential historical events (e.g., the Hamadoni flood of 2005 in Tajikistan, the Atbasar flood of 2014 in Kazakhstan, the Turkmenabat flood of 2019 in Turkmenistan, the Navoiy flood of 2019 in Uzbekistan) and for any suitable event we searched for available data and extracted the flood footprint from satellite images, when available. Unfortunately, for most of the identified cases no information on flood extent were available and the flood timing did not match with the satellite image retrieval. We could find sufficient documentation only for one flood event, occurred in Hamadoni, Tajikistan in 2005, which was the subject of the PhD Thesis of one member of the Consortium, as well as the focus of a report from JICA (Study on Natural Disaster Prevention in Pyanj River, (JICA, 2007)). All the data and the flood footprint were extracted from the JICA report.

5.3.1 Hamadoni flood probable inundation area

The Hamadoni flood occurred on the Pyanj River in Tajikistan between June and July 2005, it involved a dike breach and extensive damage to the nearby villages. Satellite data from SPOT and ASTER are only available before and significantly after the peak, as shown in Figure 30, where the upper panel shows the flood timeline an pinpoints the dates when the photographs were taken (red for Spot, blue for Aster).

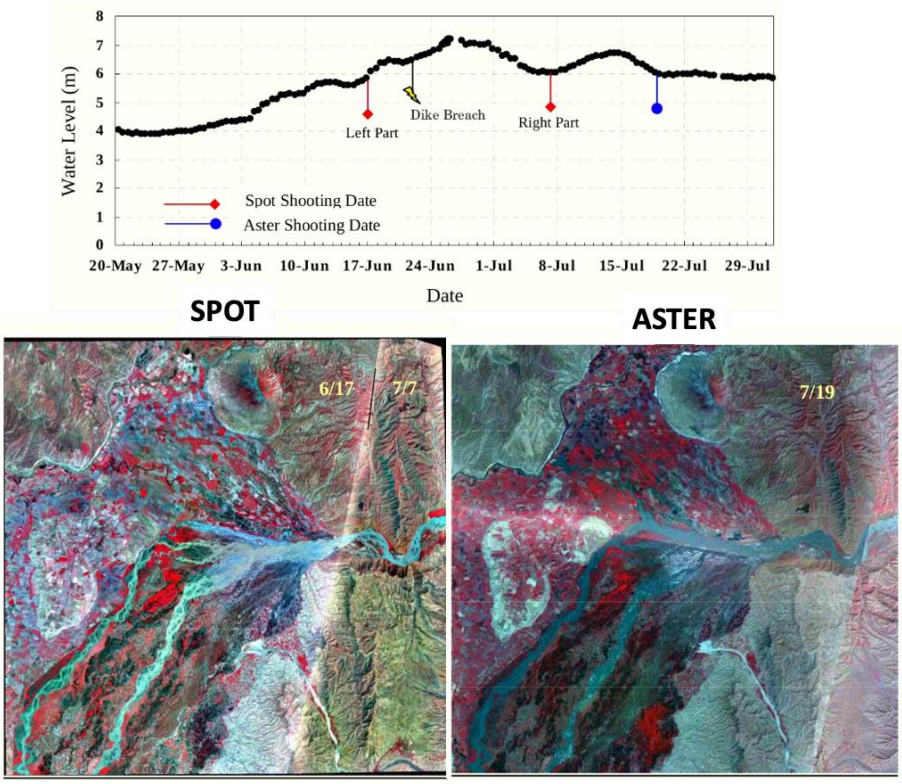


Figure 30. Satellite photographs collected around the time of the flood that covered the Hamadoni alluvial fan (source JICA, 2007).

Given this partial information, the authors of the JICA study estimate a probable inundation area shown in Figure 31.

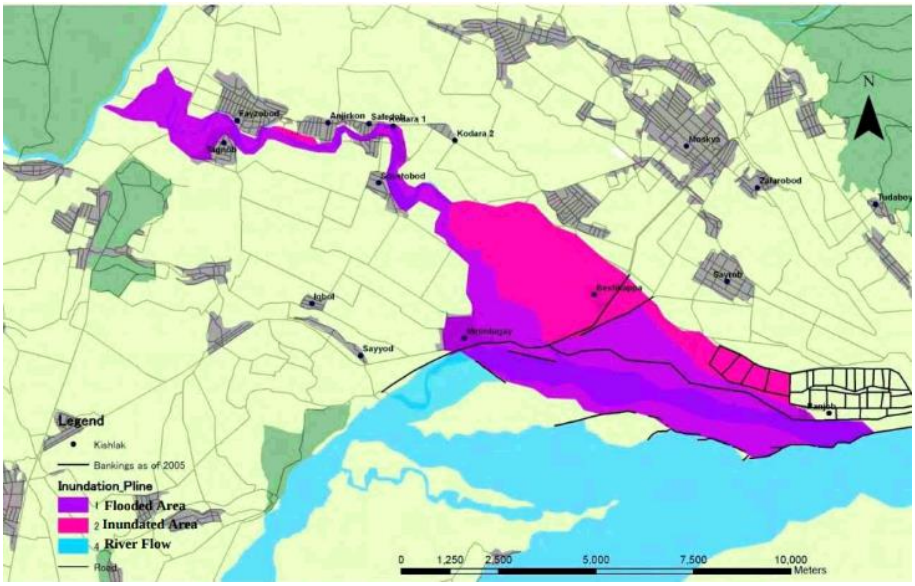


Figure 31. Probable inundation area of the Hamadoni flood event estimated in the JICA study (source JICA, 2007)

5.3.2 Discharge data and hydrograph estimation

Inlet discharge at the Chubek station was estimated in the JICA study from the data recorded at Khirmanjo (1967-2006), which is located 80 km upstream, by the peak discharge ratio. The ratio was calculated by runoff simulation of the biggest 5 historical floods with complete data (JICA, 2007). Figure 32 shows the estimated flood hydrograph from the JICA study. We used this hydrograph to run our CA2D simulation. For trial and error, we assumed a bankfull discharge of 3-year return time. This value is slightly greater than that used in our methodology because we assumed that the inundated area was protected by the hydraulic defence structures visible in Figure 31 and associated a level of protection of 3 years according to the average level of protection indicated in FLOPROS for Tajikistan. Unfortunately, more precise information on the actual level of protection and the presence and location of other defense structures and canals is not available.

5.3.3 Hamadoni flood simulation results

Figure 33 shows our flood inundated area (in yellow) compared with the satellite estimated flood map (left panel) and the JICA study simulated flood map (right panel). The latter was the result of a participated process that involved local partners right after the event occurred to discuss the construction of improved hydraulic defenses and likely had available data on the coexistent structures.

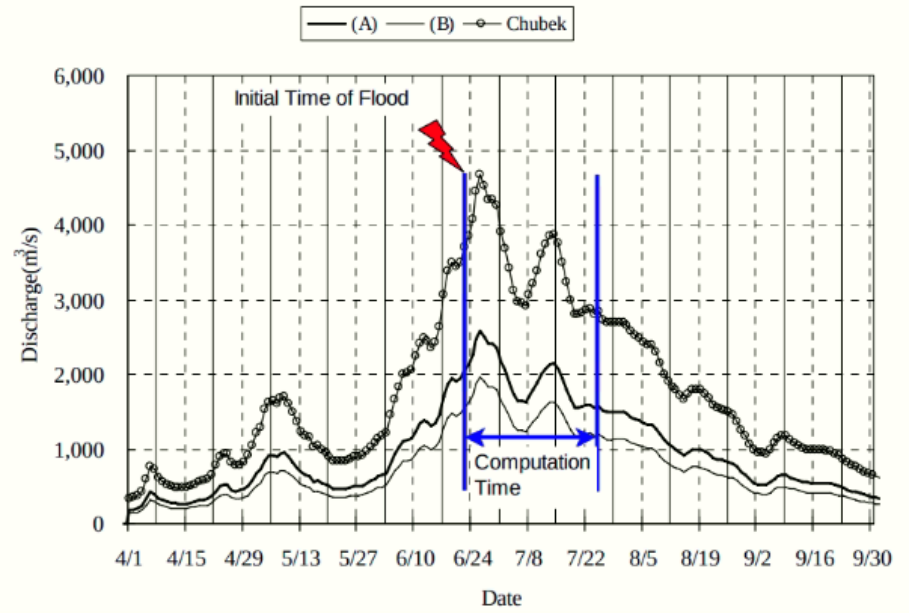


Figure 32. Estimated flood hydrograph from the JICA study. The dotted line represents the hydrograph used as input to the hydraulic model. The other lines were reported in the original figure but are not mentioned here because they are not useful for our purposes (source JICA, 2007).

Considering the lack of data and the assumptions that we had to make, we believe that the result sufficiently reproduces the historical flow behavior and inundated area. In fact, we can reasonably assume that the maximum flood extent was bigger than that obtained by the satellite images given the delay between the flow peak and the satellite image retrieval time. The time between the two events certainly allowed the furthest flooded area to drain, resulting as dry areas when the satellite retrieved the image. In addition, the assumption on the level of protection may play a role, given the clear defense structures present in the area and modeled by JICA in their analysis.

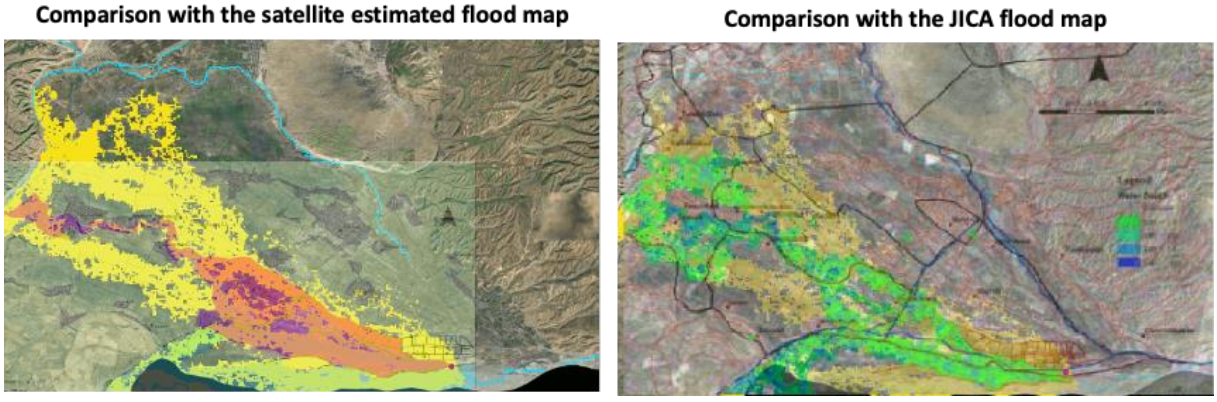


Figure 33. CA2D simulated inundated area (in yellow) compared with the satellite estimated flood map (orange and purple are in the left panel) and the JICA study simulated flood map (green area in the right panel) (source JICA, 2007).

5.4 Uncertainty

The assessment of the model uncertainty is important for a proper interpretation of the results. Uncertainty sources include:

- input data (precipitation and discharge);
- hydrological modelling;
- hydraulic modelling.

A proper uncertainty assessment is only possible if an adequate number of observations is available. In the case of the present modelling approach, observed flow data are available, thus allowing the estimation of the uncertainty of the results of the hydrological model. For this reason, the methodology followed here is based on the analysis of the hydrological model bias at gauged river sections. Such an uncertainty estimation includes errors stemming from input precipitation and temperature data and conceptual hydrological model approximations and simplifications. It does not include errors introduced by conceptual simplifications made within the hydraulic model, because observed water depth data is not available. In any case, the errors introduced by the hydraulic model are typically much smaller than the ones from the hydrological model.

The uncertainty analysis was carried out by comparing the modelled discharge extreme values distributions after bias correction with the observed ones (both obtained through a GEV fitting, see previous sections for more information). Absolute values of percent errors were computed on pairs of observed vs. simulated flows at fixed return periods (5, 10, 20, 50, 100, 200, 500, 1000 years) for every river section where observed data were available. Then, for every return period, the empirical 90% percentile of all percent errors in all gauged stations was computed. The corresponding values of percent errors were used to obtain lower and upper bounds of discharge extreme values distributions at gauged sections.

For ungauged sections, a similar procedure was put in place, but rather than comparing extreme values distributions from simulations and observations (obviously not available at ungauged sections), the comparison was done on distributions before and after the bias correction. The final result was, again, lower and upper bounds of discharge extreme values distributions.

The resulting upper and lower bound peak discharges were used to obtain upper and lower bound hazard maps for every return period through flood simulations with the CA2D model. The uncertainty in the fluvial flood simulations is therefore presented through a set of two maps for each return period, representing the bilateral 80% confidence interval of the water depth estimation in the hazard maps.

5.5 Hydraulic defenses

The role of the defensive protections is crucial in reducing fluvial flood hazard. However, availability of precise data regarding flood protection levels is very limited as most of the countries' MoES, once contacted, stressed the impossibility to disseminate such information for both confidentiality reasons and the lack of a consolidated database on flood protections.

Alternatively, we developed a strategy to derive the hydraulic protection level of the region based on the correlation between the available hydraulic protection data and the protected exposure using

the population density as a proxy of the exposure. The estimated correlation was then linearly scaled using the area protection standards provided by the FLOPROS database (Scussolini et al., 2016). However, when possible, we integrated the localized and mainly qualitative data obtained from the local partners. For example, we assumed the level of protection of the two main Kazakh cities, using geospatial information provided by the local partners.

To apply this strategy, we first identified urban agglomerates and calculated the corresponding maximum population density using the HBASE and WorldPop datasets. The former indicates the extension of urban areas. The latter provides population density over 1 km^2. Urban agglomerates were identified merging together urban areas closer than 5 km from one another (Figure 34).

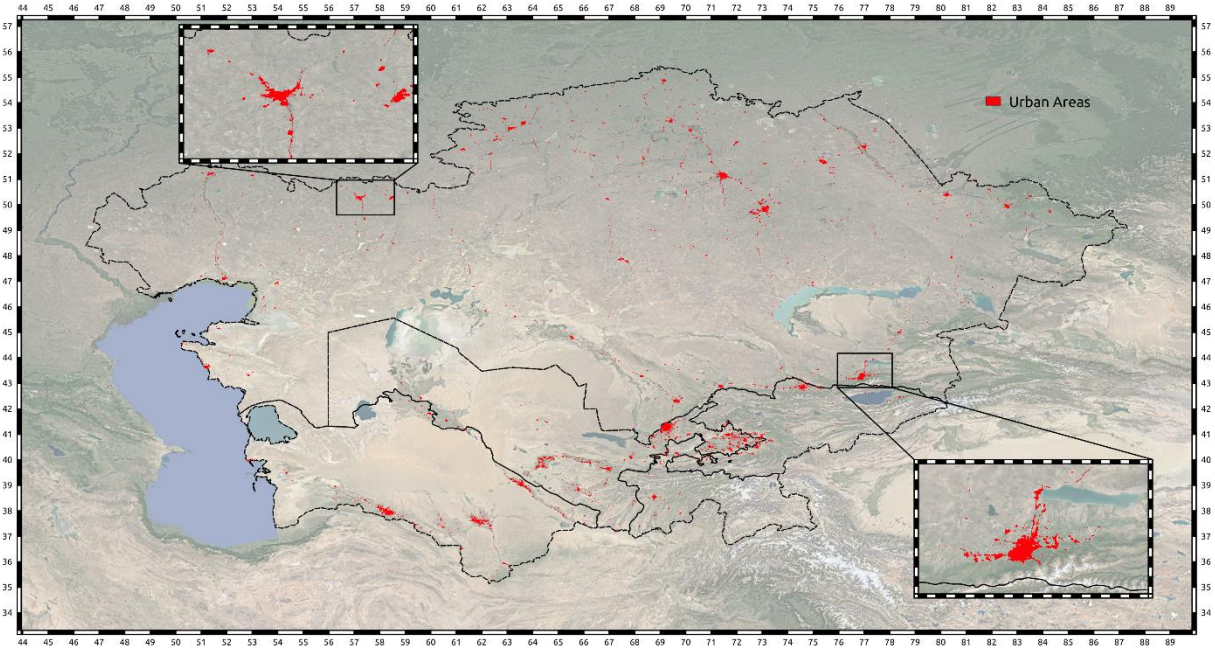


Figure 34. Urban areas derived from HBASE used to estimate the level of protection

Next, we identified the river portions afferent to each urban agglomerate by assuming that each river reach, within a distance from the agglomerate variable between 10 and 80 km depending on the accumulated area, would be affected by the relative urban level of protection.

Finally, we assigned a level of protection to each river tract afferent to the agglomerate, based on the relationship previously estimated with the population density and the area's FLOPROS protection standards. Table 6 shows the resulting hydraulic defense level of protection for each country's main cities.

Table 6. Hydraulic defense level of protection for each country's main cities (* assumed from available local data)

Country	City	Population density (pop/km²)	Level of protection (years)
Kazakhstan	Almaty	2,123	50.00*
	Nur-Sultan	545	50.00*
Kyrgyz Republic	Bishkek	2,184	15.92
	Osh	1,640	3.06
Tajikistan	Dushambe	3,931	6.91
	Khujand	1,420	2.67
Uzbekistan	Tashkent	1,953	25.13
	Samarkand	1,254	14.00
Turkmenistan	Ashgabat	727	16.32
	Turkmenabat	267	16.73

A binary effect of defenses is assumed, namely any event with a return period lower than that of the assumed level of protection would not cause any inundation outside the riverbed, while any event with a return period longer than the assumed level protection would cause an inundation equal to that obtained without any defense. This is a tenable but conservative assumption since it is based on the fact that once the level of protection of a dyke is exceeded, the dyke breaks and its protective action is not effective anymore.

The resulting hazard maps obtained with the assumed level of protections are presented for the return periods of 10, 100 and 500 years in Annex 2.

6 Realistic Scenarios

Three realistic scenarios were identified to assess potential floods that have not happened in the past but may affect the region in the future. For this reason, we asked our local experts in the consortium to provide indications on the areas where our realistic scenarios should be implemented. Specifically, we applied the following criteria: (1) flood prone area; (2) populated and built area exposed to flood risk; (3) floods happened in the area in the past.

According to the recommendation of the Ministry of Emergency Situations of the Kyrgyz Republic we simulated a 100-year flood event on the Kara-Unkur River for a flood-prone area in western Kyrgyz Republic near the city of Bazar-Korgan with a population of more than 33,300.

The resulting hazard map is shown in Figure 35. We assumed that the dyke overtop happens at the location indicated in red in the figure (72.7025° E, 41.0225° N).

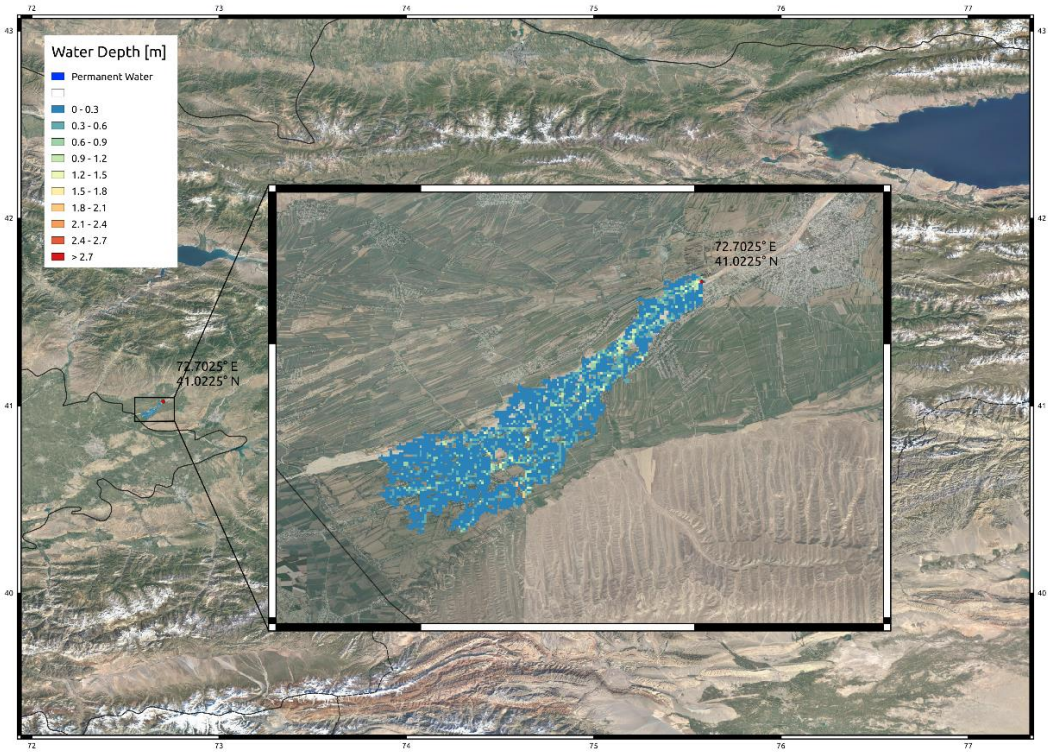


Figure 35. Realistic scenario simulation of a 100-year flood event on the Kara-Unkur River in western Kyrgyz Republic.

Our local partners from Uzbekistan, indicated Parkent city, located 47 kilometers north-east of Tashkent, and the homonymous river. Parkent is today considered one of the most flood-prone and densely populated areas, with more than 34,000 inhabitants (as of 1991) in addition to many textile and food companies.

The simulation of a 100-year flood event in the Parkent district produced the hazard map of Figure 36. We assumed that the dyke overtop happens at the location indicated in red in the figure (69.6575° E, 41.3025° N).

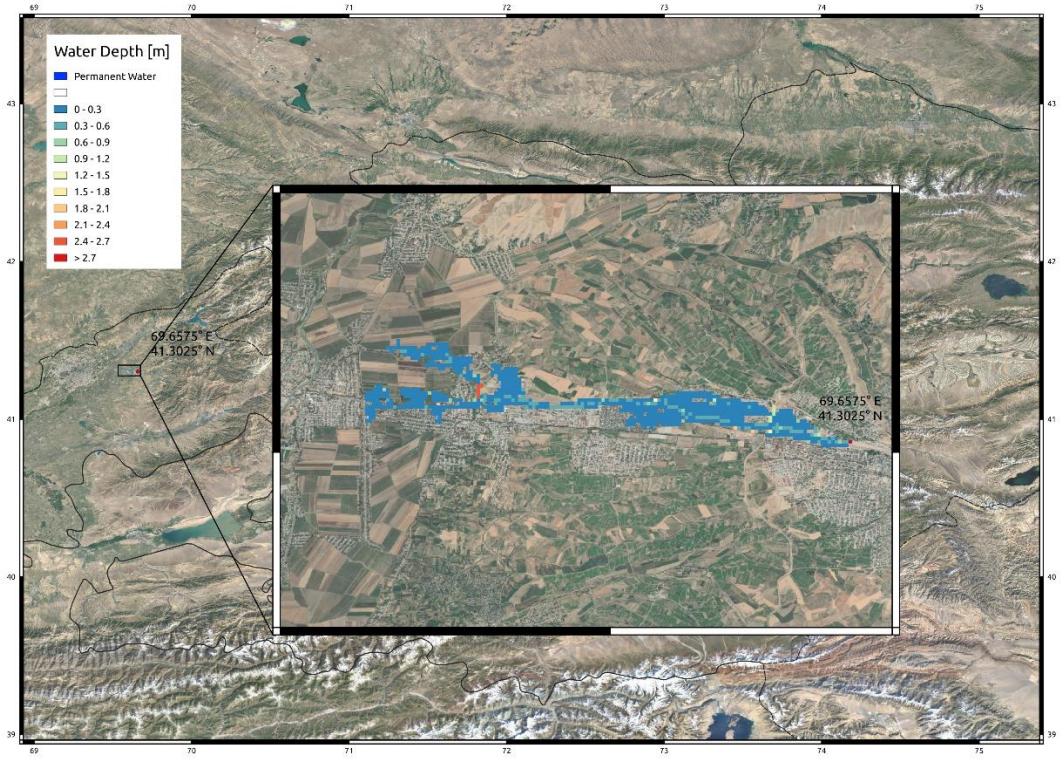


Figure 36. Realistic scenario simulation of a 100-year flood event on the Parkent river, nearby Parkent city, north-east of Tashkent in Uzbekistan.

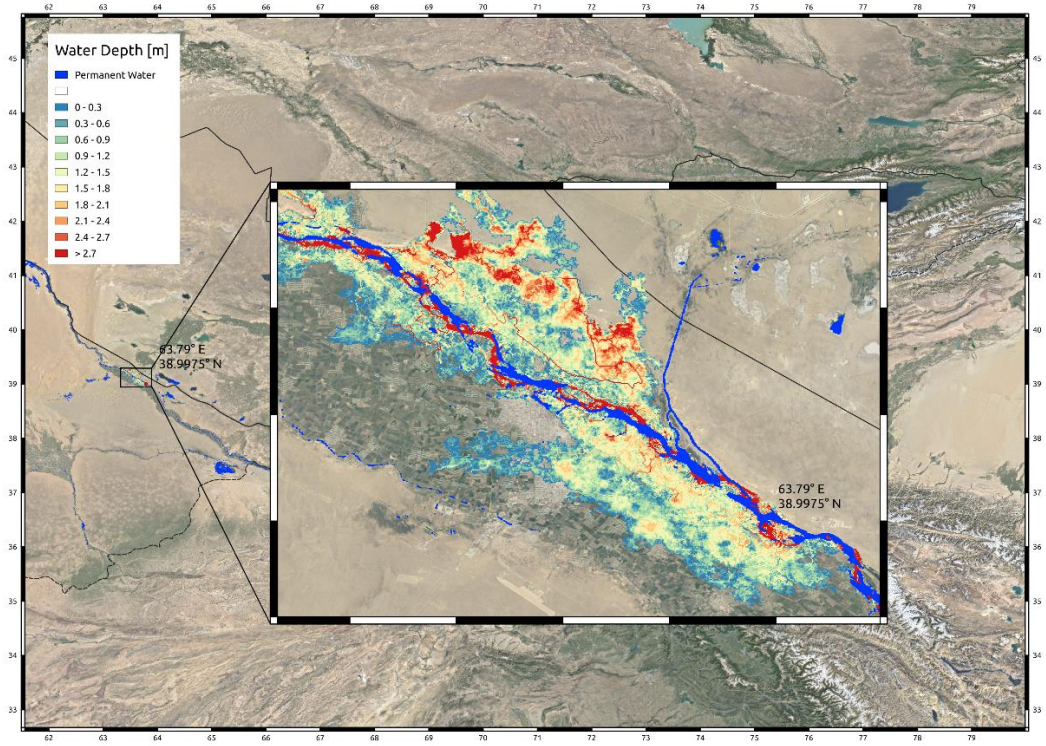


Figure 37. Realistic scenario simulation of a 100-year flood event on the Amu Darya around Turkmenabat, Turkmenistan.

Our partners in Turkmenistan recommended a focus on Turkmenabat, the second-largest city in Turkmenistan and the capital of Lebap Province on the banks of the Amu Darya River. It has a population of approximately 254,000 people (as of 2009).

The simulation of a 100-year flood event in the around Turkmenabat produced the hazard map of Figure 37. We assumed that the dyke overtop happens at the location indicated in red in the figure (63.79° E, 38.9975° N).

The flood footprints resulting from the simulation of these scenarios will be eventually used, within Task 6, to assess the flood risk over these areas in case of occurrence of such events.

7 Stochastic Catalogue

A stochastic catalogue of events is a set of plausible flood footprints that did not occur in the past but that are generated synthetically to be statistically consistent with the historical flood footprints in terms of extent, water depth and frequency of occurrence. A stochastic catalogue is necessary to obtain a reliable assessment of the losses, especially those caused by rare and very rare floods, with return periods much longer than the length of the historical record (40 years in this case). When building a stochastic catalog for risk assessment purposes the consideration of spatial correlation is fundamental, especially to obtain a reliable loss exceedance probability curve that consider the probability of having simultaneous events in different areas of the region. To build the stochastic catalogue we derived the correlation among the virtual stations based on the TOPKAPI simulated annual maximum discharge over the historical period (see Section 3) and aggregated the virtual stations into clusters based on their location, correlation, and accumulated area. Therefore, the clusters represent groups of virtual stations with high probability to have a simultaneous flood during the same rainfall event, while in virtual stations belonging to different clusters the probability to have concurrent floods is low, or even null.

The methodology followed to produce a stochastic catalogue consists of the following steps:

1. *Clustering*: virtual stations are grouped into clusters, under the assumption that flows at virtual stations in the same cluster are highly correlated random variables (Figure 38). As mentioned above, this step is based on correlations among historical simulated flows, location of the stations and accumulated areas of the stations.
2. *Cluster activation probability*: the annual probability of activation of a cluster is computed, where "activation" is defined when at least one virtual station in a given cluster exceeds the 5-year flow. This probability is based on the activation of clusters in the historical simulated flows.
3. *Activation of virtual stations within a cluster*: the average number of active virtual stations for a given year as well as its standard deviation are computed. An active station is defined as such when it exceeds the 5-year flow. These values are based on the activation of clusters in the historical simulated flows.
4. *Generation of the stochastic catalogue*: based on all the analysis above (clusters, annual activation probability, average number of active stations) and on the hazard curves at each virtual station, a stochastic catalogue is produced, with equivalent duration of 10,000 years, where every year consists in an annual flood footprint, i.e., a map where each pixel represents the maximum water depth during a given year.

The complete procedure is described in the following paragraphs.

7.1 Clustering

We first took 40 years of simulated discharge (1981-2020) and computed the annual maxima for each TOPKAPI station.

Each station was then associated to a cluster based on the annual maximum discharge correlation, location and flow accumulated area. In this phase the following assumptions were made:

- Catchments are independent; therefore, a cluster cannot include virtual stations belonging to different catchments.
- Catchments with area smaller than 5,000 km^2 become independent clusters.

The clusters were formed adopting a hierarchical clustering based on a condensed distance matrix (Johnson, 1967; Murtagh and Contreras, 2017, 2012). Hierarchical clustering (also called hierarchical cluster analysis or HCA) is an unsupervised technique to partition data into homogeneous groups called clusters. In order to decide which elements should be combined, a measure of dissimilarity between sets of observation is required. Typically, this is achieved by defining a distance metric (a measure of distance between pairwise observations) and a linkage criterion which defines from which cluster point the distance is calculated.

We built the distance matrix as a weighted combination of the correlation matrix and the Gower's distance matrix (Gower, 1971) computed from the station's location (coordinates) and flow accumulated area. As linkage criterion we selected the Ward method which minimizes the total within-cluster variance (Ward, 1963).

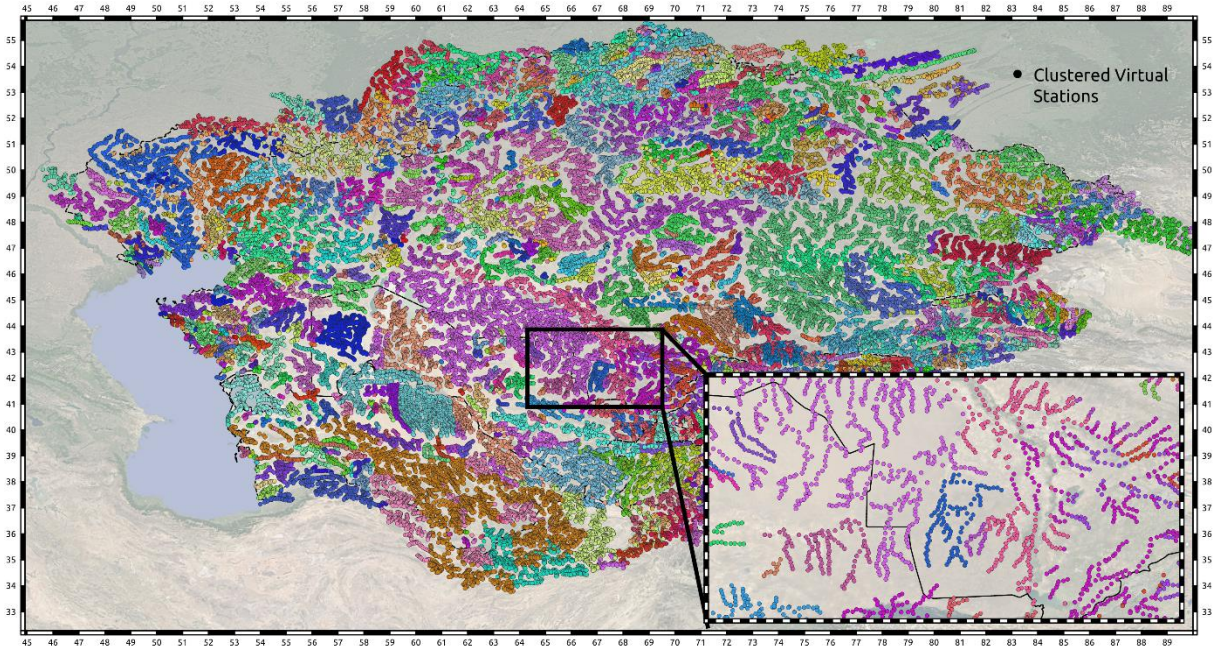


Figure 38. Map of clustered virtual stations, virtual stations with the same colour belong to the same cluster

7.2 Cluster activation probability and number of active stations

To estimate the annual cluster activation frequency, we assume that any station in a certain cluster is active for a given year if its annual maximum discharge value is larger than its relative 1-in-5 years flow peak value. A cluster is considered active if at least one of its stations is active at any given year.

For each cluster we extract two features:

- The cluster activation frequency (f_a) which indicates the probability that at least one station in the cluster is active in any given year.

- The mean and standard deviation of the number of active stations in an active cluster. From these two moments, we estimate the parameters' values of a truncated normal probability distribution P_a, describing the number of active stations in an active cluster.

7.3 Stochastic catalogue generation

For each year *y* of the 10,000-year stochastic catalogue, the following algorithm was implemented:

For each cluster *c:*

1. We define if the cluster *c* was active based on its frequency of activation
2. If *c* is active, we define the number of active stations in the cluster by drawing a value from the truncated normal probability distribution P_a that was previously estimated.
3. We randomly associate each active station in the cluster to a return period longer than 5 years.
4. For each active station in the cluster, its flood footprint is obtained simulating the CA2D model with an input discharge corresponding to the associated return period.

All the flood footprints generated for a given year *y* from all the active stations in the region represent the maximum water depth occurred at any grid cell of the domain during year *y*.

Applying these steps for each of the 10,000 years will result in 10,000 annual flood maps that constitute the hazard-consistent stochastic catalogue.

The consistency of the stochastic catalog with the hazard maps has been verified through two different analyses:

- Verification of activation frequency of each virtual station
- Comparison of the hazard curves derived from the stochastic catalog with the hazard values extracted from the hazard maps

The verification of the activation frequency is shown in Figure 39, where the annual frequency of activation, computed as the inverse of the annual occurrence rate, for different return periods are depicted. The theoretical value for the return period P represents the annual probability of having an event with a return period equal or greater than P, it is computed as 1/P. The simulated frequency is derived from the stochastic catalog considering the actual number of occurrences of events with a return period equal or greater than P during the 10.000 years of simulations. The values reported in Figure 39 represent, for several return periods, the average activation frequency of all the 78,000 virtual stations and show a very good agreement with the theoretical behavior.

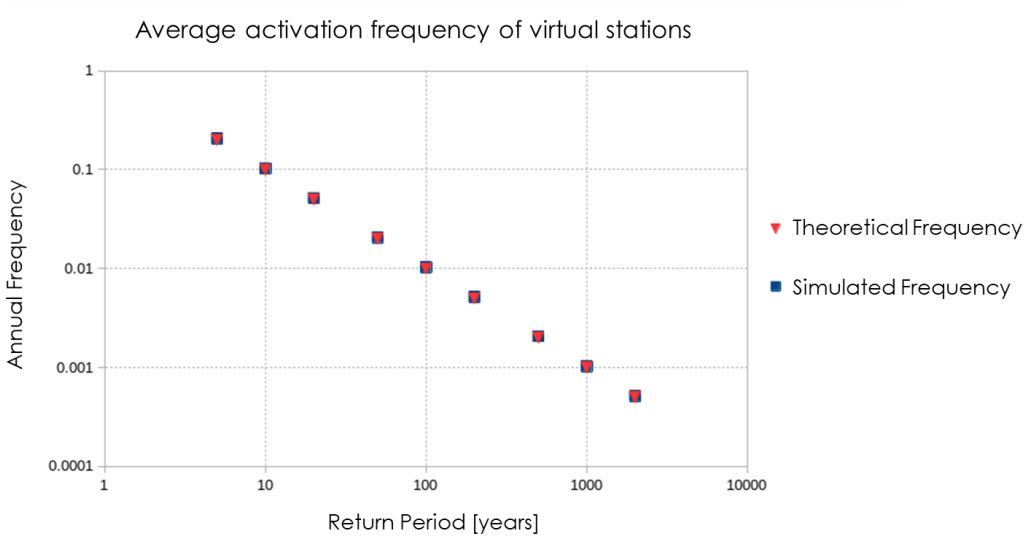


Figure 39. Average activation frequency of the 78.000 virtual stations. The chart represents the expected frequency of activation with a return period above a certain value in both the theoretical and simulated cases.

Regarding the comparison of the hazard curves, we randomly selected 120 locations with flood hazard greater than 0 within the entire domain and calculated the bias between the hazard derived from the stochastic catalogue and the hazard extracted from the hazard maps for each return period. Figure 40 shows the bias for the predefined return periods used to generate the hazard maps including the standard deviation, while Figure 41 shows the comparison over 4 locations in different countries. The bias ranges between -2.0 and +3.5 cm, showing a good agreement between the stochastic catalogue and the hazard maps.

Finally, Figure 42 shows an example of one annual flood footprint taken from the 10,000 annual maps generated for the undefended fluvial flood stochastic catalogue.

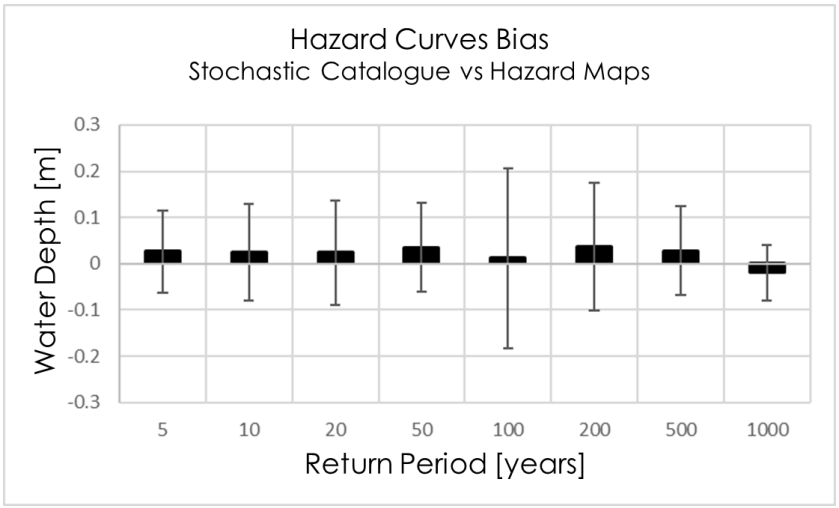


Figure 40. Bias between hazard curves extracted from the stochastic catalogue and hazard values extracted from the hazard maps, computed over 120 points randomly extracted over the entire domain

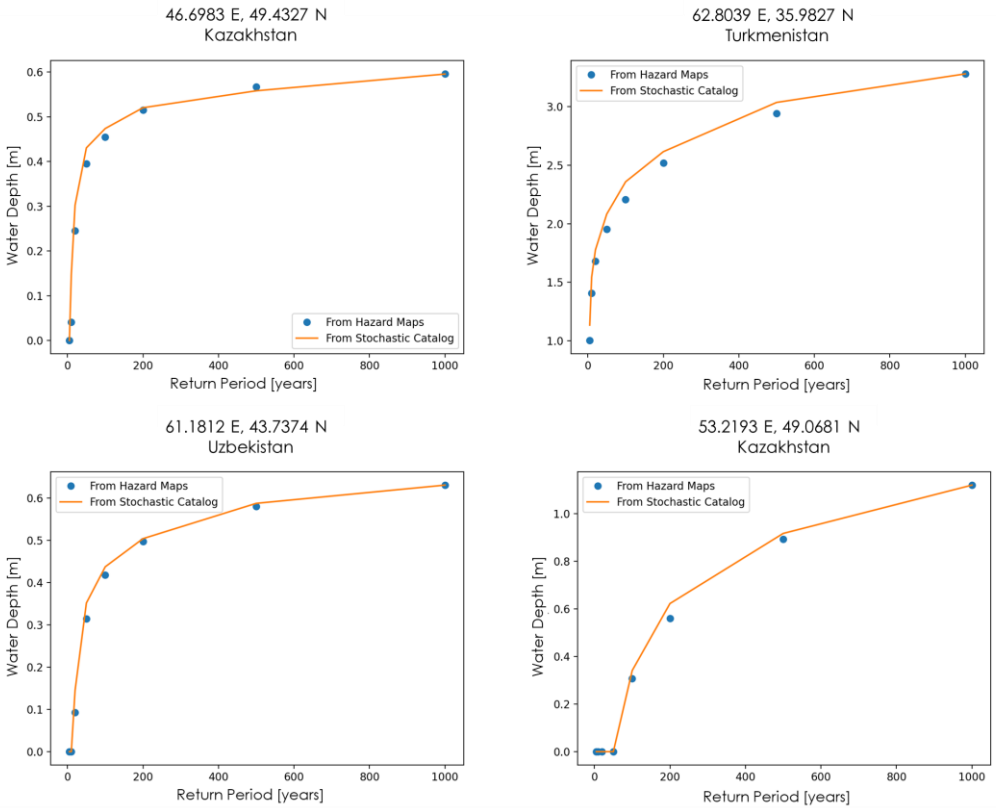


Figure 41. Comparison of hazard curves extracted from the stochastic catalogue with the hazard values extracted from the hazard maps for the undefended case.

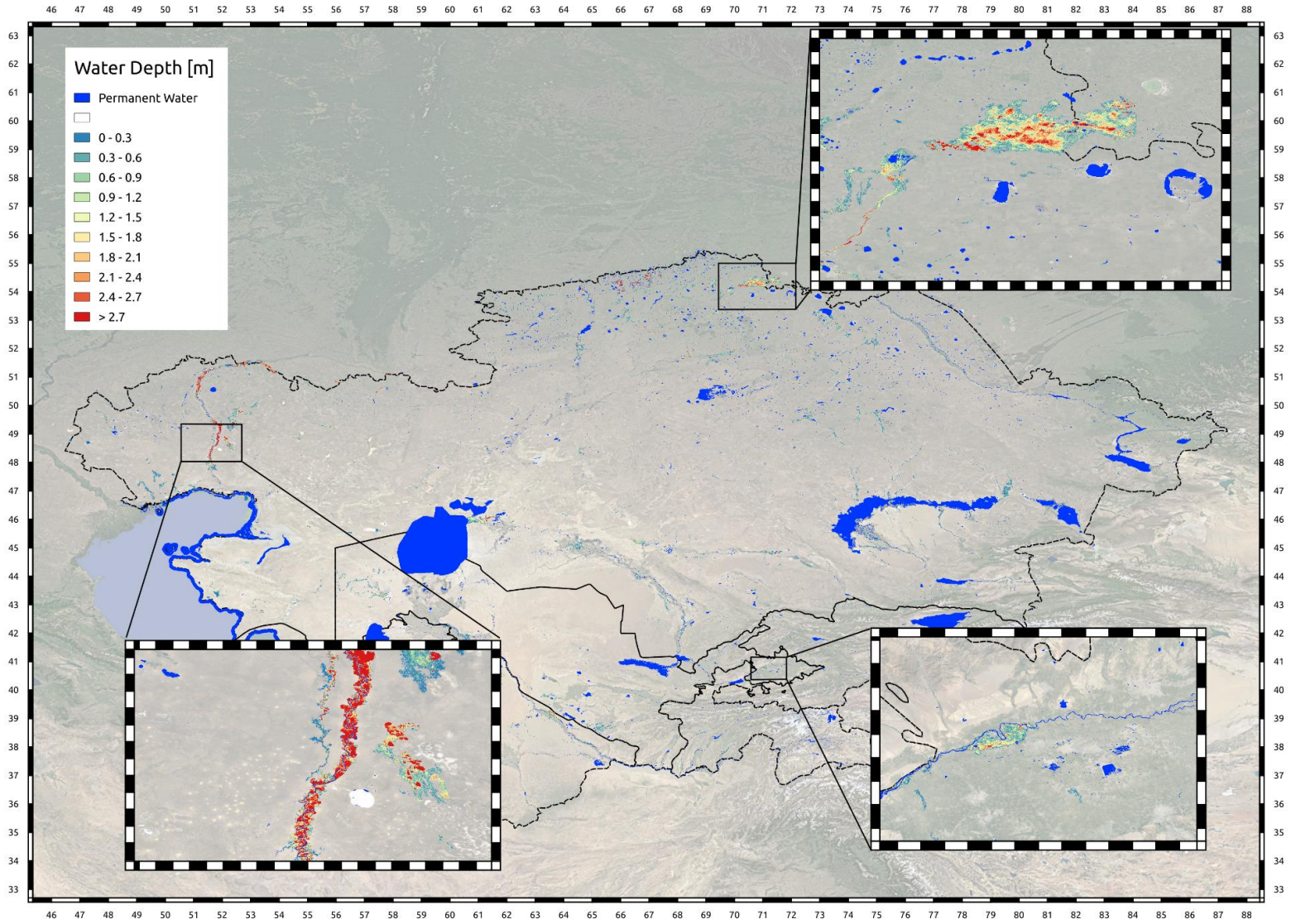


Figure 42. Example of an annual flood footprint extracted from the 10.000 years composing the undefended fluvial flood stochastic catalogue

8 Climate Change scenario

Central Asia experiences very large temperature fluctuations, which are expected to increase in severity and frequency due to climate change. The effects of a changing climate are already visible today, with noticeably higher temperatures and an increase in extreme weather events frequency and severity in recent decades (Hydrometeorological Service under the Cabinet of Ministers of the Republic of Uzbekistan, 2009; Ministry of Environment Protection Kazakhstan, 2009; State Agency for Hydrometeorology of Tajikistan, 2009; State Agency on Environment Protection and Forestry under the Government of the Kyrgyz Republic, 2009). Over the course of 80 years, all countries have seen an increase in annual average temperatures that significantly exceeds global trends. The countries of Kazakhstan, Kyrgyz Republic, and Tajikistan experienced the highest temperature increase in the winter period, while in Turkmenistan and Uzbekistan the most significant changes were observed in the summer and autumn months. Even if a regional clear trend in precipitation was not observed, precipitation variability and the frequency of intense rainfall events have consistently increased in all countries, especially in the mountain areas where two-thirds of the region's population is concentrated[1].

Higher temperatures triggered by climate change are expected to significantly affect the hydrological cycle in the region, especially for what concerns both the snow accumulation/melting, which represents one of the main drivers of fluvial floods in the area, and the evapotranspiration processes, which may reduce water availability in the dry periods and increase the water demand for irrigation.

Our goal is to evaluate the change in fluvial flood hazard for a 2080 scenario. The analysis is based on climate change scenarios over the region available through the data repositories of the Copernicus climate data store (Alos et al., 2019) and the Coordinated Regional Climate Downscaling Experiment (CORDEX) project database (Giorgi et al., 2009).

8.1 Literature review and selection of a climate scenario

We need to identify a suitable scenario for our analysis, the selection should include: a Global Circulation Model, a Regional Climate Model (RCM) i.e., a downscaling method to reproduce and project fine-scale climate information from GCMs, and a Representative Concentration Pathways (RCP) i.e., one of the greenhouse gas concentration trajectories adopted by the IPCC.

After completing a literature review on climate change in Central Asia (e.g., He et al., 2021; Liu et al., 2020; Ozturk et al., 2017; Top et al., 2021; Yao et al., 2021; Zhu et al., 2020), we focused on the results presented in the paper by Ozturk et al. who used the Regional Climate Model (RCM) RegCM4.3.5 to project changes in temperature and precipitation in Central Asia within the CORDEX framework. The study included projections from three periods: 2011-2040, 2041-2070 and 2071-2100 and used the historical period 1981-2000 as reference. The adopted RCM was

[1] First and second National Communication on Climate Change under the UNFCCC (see previous reference)

RegCM4.3.5 (RCM), a hydrostatic nature regional climate model. The analysis considered two different GCMs: MPI Earth-System Model (ESM), running on medium resolution (MR) grid, and HadGEM2-ES, a coupled Earth System Model developed for CMIP5 centennial simulations. We decided to select RCP 4.5 which is described by the IPCC as an intermediate Representative Concentration Pathways scenario: in RCP 4.5 emissions are expected to peak around 2040, and then decline (see Figure 43). RCP 4.5 is more likely than not to result in global temperature rise between 2°C and 3°C by 2100.

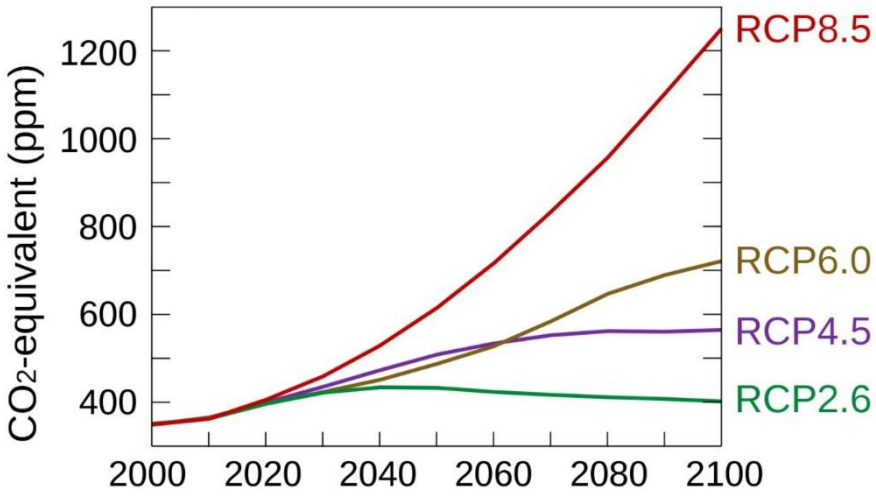


Figure 43. IPCC Representative Concentration Pathways

Figure 44 and Figure 45 show the geographical distribution patterns of changes in projected mean air temperatures and total precipitation, respectively, for Central Asia in the period 2071-2100 (Ozturk et al., 2017) for the selected RCP 4.5 and the two considered GCMs: the HadGEM2-ES (upper panel) and the MPI-ESM-MR (lower panel). For both models the warming trend is most observable in the north of the domain and mean air temperatures will increase more in the far future than in the mid-future and near future periods, with air temperatures varying from 3 °C up to 8 °C on average for the period of 2071-2100. A decrease in precipitation is also expected for the region.

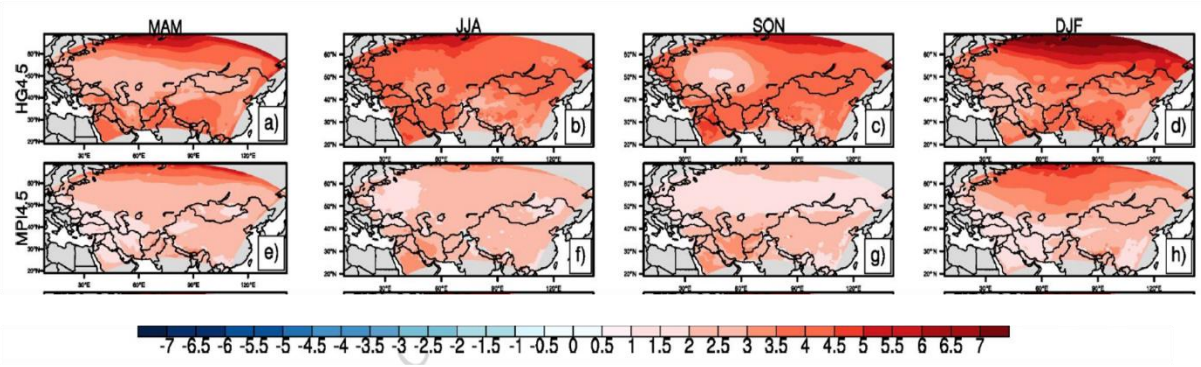


Figure 44 Geographical distribution patterns of changes in projected mean air temperatures (°C) over Central Asia from (source Ozturk et al. 2017)

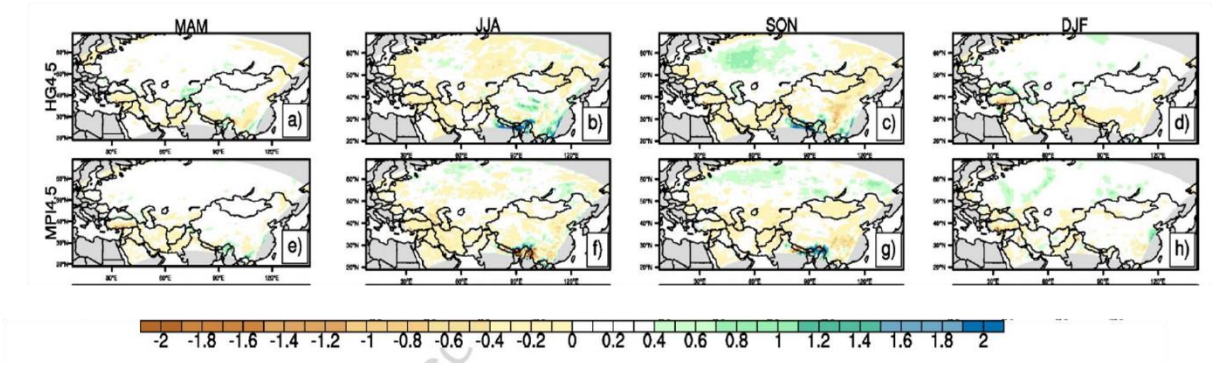


Figure 45. Geographical distribution patterns of changes in projected total precipitation (mm/day) over Central Asia (source: from Ozturk et al. 2017)

Results obtained from the regional model driven by the MPI-ESM-MR global model are milder than results of the regional model driven by the HadGEM2-ES global model. For this reason, we select the MPI-ESM-MR GCM for our climate change scenario simulation.

8.2 Methodology

To run the climate change scenario simulation, we first adopted change factors to the ERA5-Land precipitation and temperature based on the pdf comparison of the current climate and 1971-2100 projection. We used a pdf matching technique to change the distributions of the current ERA5-Land variables. This technique was preferred to a simple identification of constant correction factors and allows to account for changes in extreme events. Eventually, the entire modeling chain composed of hydrological modeling (TOPKAPI), extreme values analysis and hydraulic modeling (CA2D) was fed with the modified ERA5-Land derived meteorological input data (precipitation and temperature) in order to derive the hazard maps for the 2080 scenario.

The extreme values correction factors obtained during the extreme value analysis (Section 4) were applied to the hydrological model outputs to keep consistency with the current climate flood hazard assessment.

The resulting hazard maps are presented for the return periods of 10, 100 and 500 years in Annex 3.

8.3 Results Discussion

The impact of the adopted climate change scenario on the hazard maps is visible for most return periods.

What we observe is that regional warming produces a reduction in snow accumulation everywhere, but its impact is particularly noticeable at the southern latitudes. This effect translates into a sizeable reduction of both normal and extreme flows in the most Southern basins (i.e., the upper Amu Darya), in particular in spring/summer, when rivers are fed by snowmelt. This triggers a proportional reduction of the flood hazard. In the northern part of the domain, warming still induces faster and earlier snowmelt, but in this case extreme flows show an increase, triggered by an increase in extreme rainfall event, which appear to become more intense in the future scenarios

considered in this modelling framework. This translates into a moderate increase in flood hazard. Overall, the adopted climate change scenario is expected to produce a slight reduction in flood hazard at the country scale in Central Asia, especially in Turkmenistan and Tajikistan.

9 Pluvial Flood

The modelling approach described in the previous sections covers fluvial floods, i.e., inundations caused by medium to large-sized rivers, while localized heavy precipitation bursts might not be accurately reproduced. This is a crucial issue for several areas of the region, which are characterized by short and intense floods, such as the mountainous areas of Tajikistan and Kyrgyzstan, that may trigger catastrophic mudslides as often observed in the past.

In order to assess pluvial flood hazard in Central Asia, we fed the CA2D model with extreme precipitation events based on the Intensity-duration-Frequency (IDF) curves derived from ground stations observations and remotely sensed data. Spatial correlation is taken into account through the definition of a cross-correlation matrix estimated on the historical rainfall intensity time series. The infiltration rate is also taken into account as a function of the land use and soil type classes as well as the presence of urban areas. This ensures that the model is calibrated and validated specifically for the study area. Excess of rainfall is routed using the model propagation equations over selected areas and used to calibrate a geomorphological index in order to derive water depths over the entire model domain.

9.1 Data processing

9.1.1 Derivation of Intensity-Duration-Frequency curves

The first step to assess pluvial hazard was the derivation of the Intensity-duration-Frequency (IDF) curves over the entire region. The IDF curves represent the distribution of the extreme precipitation for different durations and have been derived according to the ERA5-Land precipitation data for each grid cell and for the durations of 3, 6 and 9 hours, which represent the most critical durations for pluvial flood formation. In order to derive the IDF curves, the ERA5-Land data were spatially aggregated to tiles of 0.4°x0.4° resolution, and for each tile the annual maximum values of precipitation corresponding to the above-mentioned durations were computed and fitted with a Gumbel distribution.

As already mentioned in Section 2.2, precipitation data from ERA5-Land represent volumes averaged over the grid cells, whose size is about 100 km^2. The spatial averaged precipitation maintains the average volumes of precipitation in the area and is suitable for hydrological modelling purposes and for fluvial flood hazard assessment, but it unavoidably leads to underestimating the local peaks of rainfall that drive the pluvial hazard. For this reason, before deriving the IDF curves, a comparison between ERA5-Land and the KNMI Climate Explorer (CE) precipitation data was performed to evaluate the magnitude of the gridded data bias with respect to the point station data. Figure 46 shows the spatial distribution of the CE stations and the tiles used to setup hydraulic simulations for pluvial flood hazard assessment.

The CE precipitation data were obtained in daily values for all available stations in each country of Central Asia, for the period 1981-2020 in order to cover the ERA5-Land available data time frame. Only the stations with more than 300 days of data per year and with at least 10 years of observations were selected for the comparison. Specifically, we focused on the extreme values of the annual maximum precipitation.

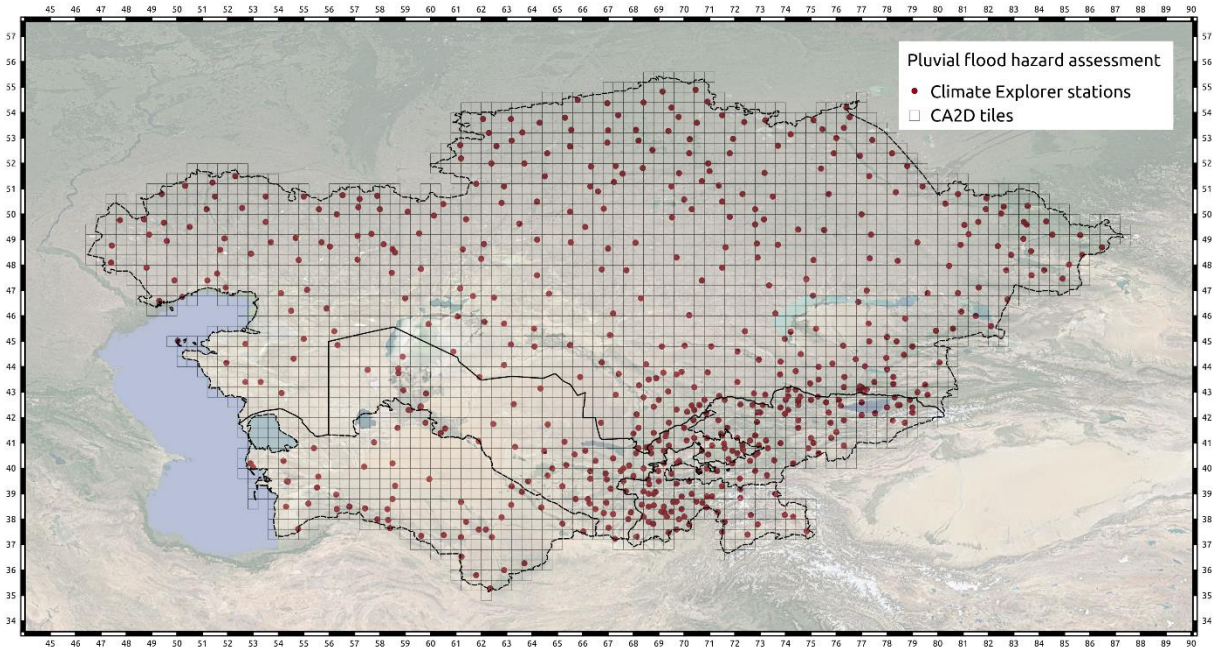


Figure 46. KNMI Climate Explorer stations and 0.4°x0.4° tiles used to simulate pluvial flood hazard

For each country, the percent bias of the highest 10 annual maximum precipitation values from each CE station and its corresponding ERA5-Land tile was computed and used to correct the extreme precipitation from ERA5-Land (Figure 47). As expected, we found that the highest bias was associated to the highest value of the series of each station, while it decreased to a very low value for the tenth annual maxima. Based on these results, we corrected the highest ten values of the historical series of the ERA5-Land annual maxima precipitation of each 0.4°x0.4° tile by applying a correction coefficient that decreases with the ordering positioning. This coefficient was derived by the comparison between ERA5-Land and CE data.

Figure 48 shows the percent bias obtained by comparing ERA5-Land and CE data over each country. Figure 48 shows a comparison between the original and the corrected ERA5-Land annual maxima series for five selected locations across the region.

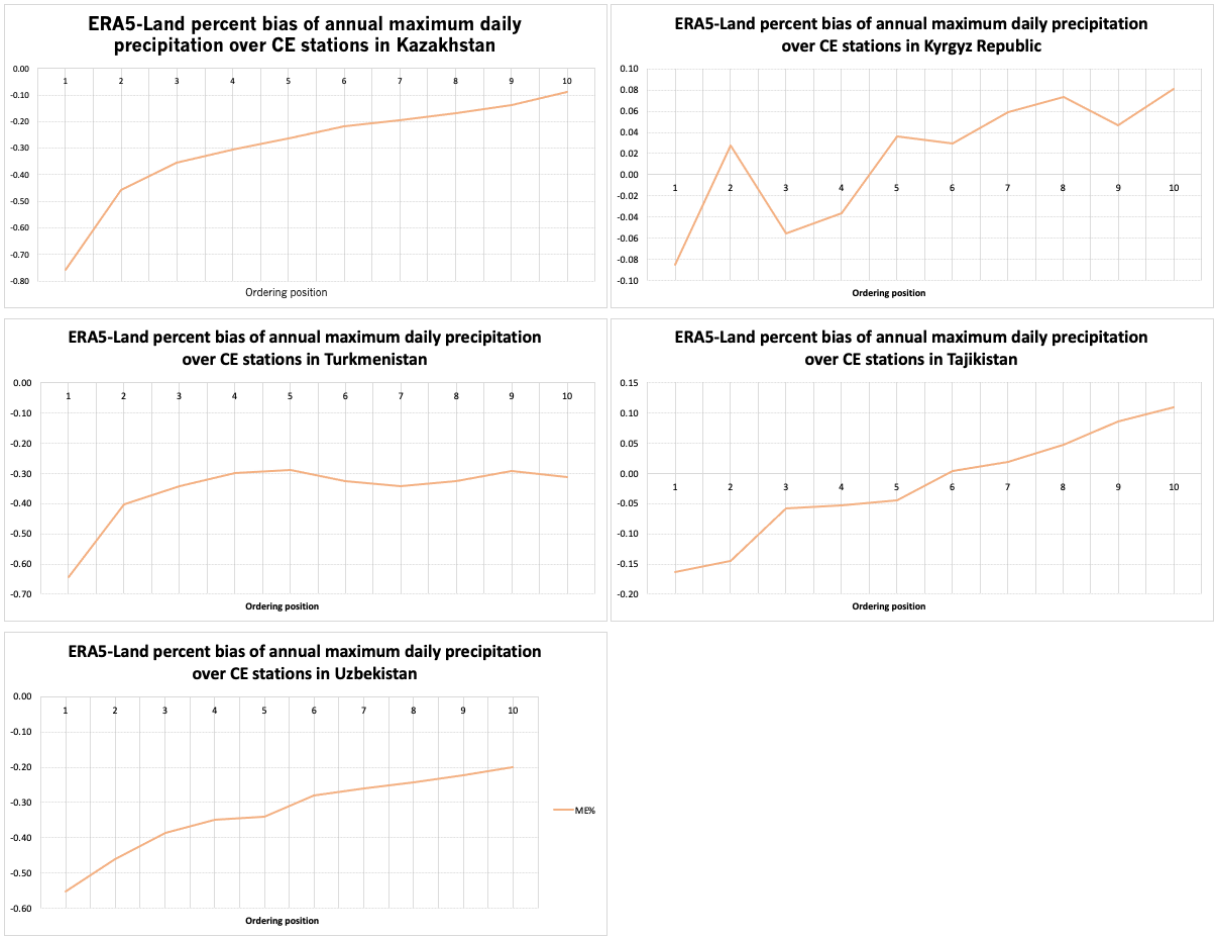


Figure 47. Percent bias of the ten highest annual maximum precipitation values from ERA5-Land compared to the CE stations data for the five target countries

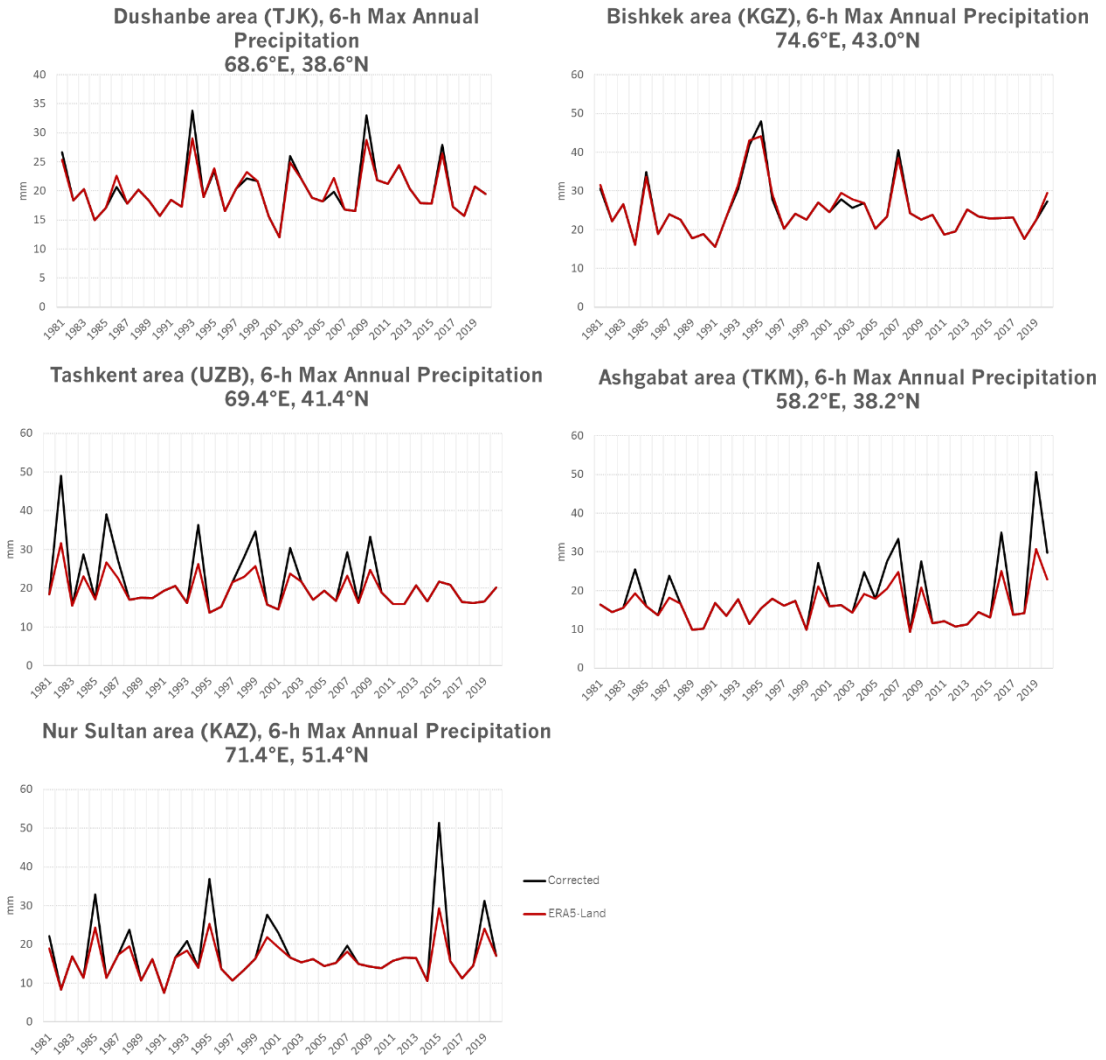


Figure 48. Comparison between original ERA5-Land and corrected annual maximum precipitation series for several locations in Central Asia

9.2 Hydraulic model simulations

The CA2D model was used to simulate the effect of the extreme precipitation obtained correcting the ERA5-Land data as described in the previous section. Similarly to the fluvial hazard model configuration, the simulation domain is discretized into tiles. Each 0.4° tile was assumed to be affected by the same precipitation constant over time for the durations mentioned above (3, 6 and 9 hours) plus a 20 percent to allow for additional routing after the end of the event. To avoid potential abrupt changes along the border of the tiles, the hydraulic model was setup for greater tiles of 0.5°x0.5°, which overlapped the surrounding tiles of 0.1°. Over the overlapping areas, the higher water depth simulated in adjacent tiles was selected.

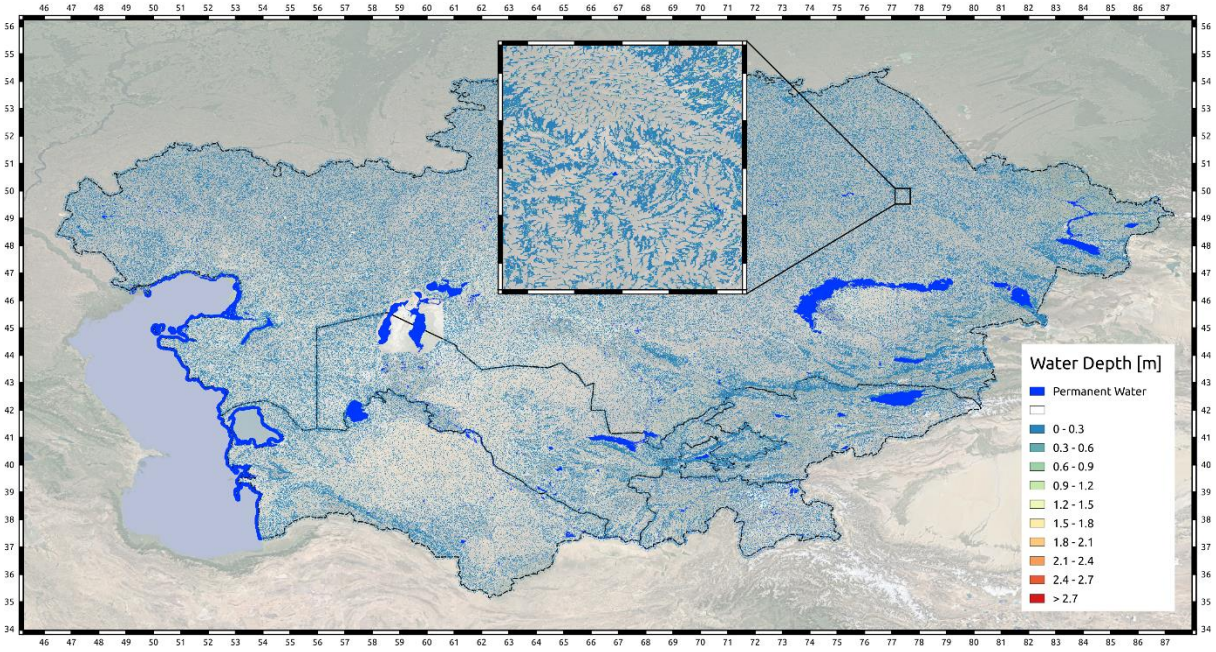


Figure 49. Zoom of a CA2D simulation output for one 0.5°x0.5° tile

The same DEM and roughness data used for fluvial flood were used for these simulations and an infiltration map was derived assuming a general infiltration rate of 30% on non-urban areas and of 5% over urban areas. The urban areas were identified based on HBASE dataset (Figure 34). Finally, pluvial flood was assumed to be caused by localized intense precipitation events and generated by accumulation of water before reaching the main drainage network. For this reason, we set a threshold of accumulated area above which the water was supposed to reach the main river network and not being anymore a threat for pluvial flood. This threshold was set to 30 km^2.

9.3 Geomorphological model calibration to reduce computational effort

To reduce the computational effort required for a fully physically based pluvial flood simulation over such a wide region, we took advantage of the geomorphological characteristics of the terrain, which is the main driver of potential pluvial floods together with the precipitation intensity and duration. In fact, we first identified a set of relevant areas, evenly distributed across the domain, for which we ran a full CA2D pluvial simulation. Then, we calibrated over these areas a conceptual model based on a geomorphological index, the Compound Topographic Index (CTI) (Figure 50), to produce the hazard maps over the rest of the region. The CTI was developed by Beven and Kirkby within the runoff model TOPMODEL (Beven and Kirkby, 1979).

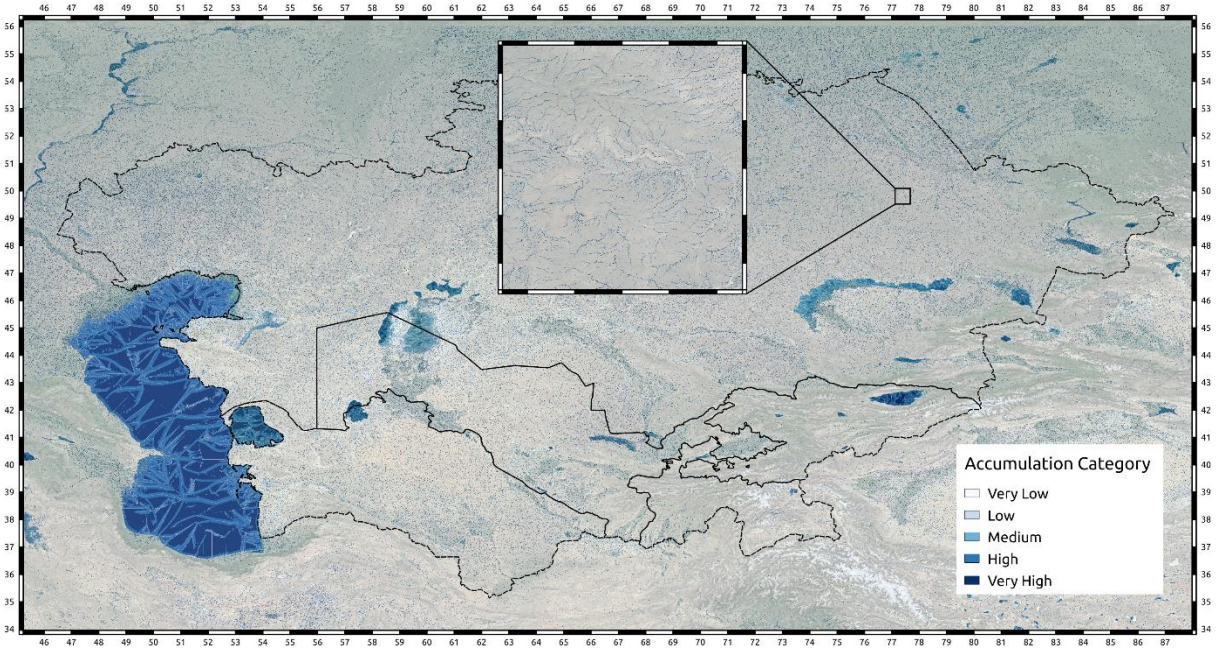


Figure 50. Compound Topographic Index aggregated in categories to show its ability to identify area of water accumulation based on a simple geomorphological relation.

The CTI, also known as the Topographic Wetness Index (TWI), is a measure of the extent of flow accumulation at any given point of the topographic surface and combines information on the flow accumulated area (A) and slope (S) and is defined as follows:

$$CTI = \ln\left({}^{A}/_{\tan(S)}\right)$$

As the flow accumulated area increases and slope gradient decreases, the CTI increases. The CTI was therefore used as a weight to distribute the volume of precipitation felt upstream each MERIT DEM grid cell and to derive the water depth caused by the precipitation events with different return periods (5, 10, 20, 50, 100, 200, 500 and 1000 years).

The resulting hazard maps are presented for the return periods of 10, 100 and 500 years in Annex 4.

10 Concluding Remarks

10.1 Strengths and limitations

The results presented herein constitute the first attempt to date at assessing flood hazard in Central Asia at a regional scale using a uniform methodology and a homogenized dataset that includes both global and local data sources. The cooperation with local partners was productive and gave the consortium access to a set of unprecedented local datasets and information as well as technical advice. The partnership with local governmental institutions and authorities also offers the potential for integration with national models. Therefore, the main strength of this project is the combination of state-of-the-art global flood modeling techniques with an unprecedented use of local data and knowledge that allowed for a sound hazard assessment over such a wide area.

The proposed modelling approach is also potentially suitable for the implementation of Disaster Risk Financing applications based on parametric triggers (e.g., CCRIF for the Caribbean and Central America regions, ARC for the continent of Africa).

The main limitation remains the general lack of water depth observations for validating the hydraulic model output as well as specific data on hydraulic defense structures, their location, level of protection and maintenance. We understand the impossibility of sharing sensible information and believe we made the most accurate assumptions given the available data. However, we recognize that the presence of dikes can have a non-negligible localized effect on flood hazard.

As for pluvial hazard, we included the effect of urban drainage but we had to assume the urban and non-urban infiltration rates (5% and 30% respectively) given the lack of data for the region. The optimal approach to assess pluvial hazard would have been to simulate 10,000 stochastic years, however, this solution would be impractical at this scale in terms of computation time and resources

10.2 Recommendations on users and applications

The accuracy of any model result is affected by the input data quality and availability and the model assumptions. In our application, the pluvial flood hazard has greater uncertainty than the fluvial flood hazard, because there are more uncertain localized factors that can influence pluvial flooding but are not represented in our model. The same is true for defended flood hazard, whose accuracy is limited by the quality and availability of hydraulic defense structures information. 2D hydraulic models are known to perform better on large rivers and, therefore, fluvial flood hazard maps are expected to be more accurate in large floodplains than in mountainous headwater areas. Because of these limitations, caution should be used for the interpretation of the hazard estimates at the very local scale, also considering the spatial resolution of 90m that does not allow for a detailed assessment of flood footprints at the local scale. The use of these data is, therefore, not recommended for detailed planning or operational decision as well as for engineering-level analysis of specific assets. That been said, we are fully convinced the hazard estimates computed herein are appropriate for assessing regional flood risk, which is the final objective of this study.

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Appendix A - List of acronyms

ARC: African Risk Capacity

CA: Cellular Automata

CAC DRMI: Central Asia and Caucasus Disaster Risk Management Initiative

CCRIF: the Caribbean Catastrophe Risk Insurance Facility

CE: Climate Explorer

CFL: Courant–Friedrichs–Lewy

CIMA-UNEP: Centro Internazionale in Monitoraggio Ambientale and United Nations Environment Program

CMIP5: Coupled Model Intercomparison Project phase 5

CORDEX: Coordinated Regional Climate Downscaling Experiment

CTI: Compound Topographic Index

DEM: Digital Elevation Model

DRFI: Disaster Risk Financing and Insurance

DRM: Disaster Risk Management

DRR: Disaster Risk Reduction

ECMWF: European Centre for Medium-Range Weather Forecasts

ESM: Earth-System Model

EV: Extreme Value

FAO: Food and Agriculture Organization

FLOPROS: FLOod PROtection Standards

G1WBM: Global 1 arc-second Water Body Map

GCM: Global Circulation Model

GEV: Generalized Extreme Value distribution

GFDRR: Global Facility for Disaster Reduction and Recovery

GloFAS: Global Flood Awareness System

GLOFRIS: Global Flood Risk with Image Scenarios

GRanD: Global Reservoir and Dam Database

GRDC: Global Runoff Data Centre

GSWO: Global Surface Water Occurrence

HBASE: Human Built-up And Settlement Extent

HCA: hierarchical cluster analysis

HSWD: Harmonized World Soil Database

IDF: Intensity-duration-Frequency

IPCC: Intergovernmental Panel on Climate Change

IS: Institute of Seismology

ISNASKR: Institute of Seismology of Kyrgyz Republic

IWPHE: Institute of Water Problems, Hydropower Engineering and Ecology

JICA: Japan International Cooperation Agency

JRC: Joint Research Centre

KNMI: Royal Dutch Meteorological Institute

LAEA: Lambert Azimuthal Equal Area

LMOM: Method of L Moments

MERIT: Multi-Error-Removed Improved-Terrain DEM

MES: Ministry of Emergency Situations of the Kyrgyz Republic

MLE: Maximum Likelihood Estimation

MOM: Method of Moments

MPI: Max Planck Institut

MR: Medium Resolution

NGCC: National Geomatics Center of China

NW: Northwest

RCM: Regional Climate Model

RCP: Representative Concentration Pathways

RED: Risk Engineering + Development

SFRARR: Strengthening Financial Resilience and Accelerating Risk Reduction in Central Asia

TWI: Topographic Wetness Index

UN: United Nations

UNDRR: United Nations Office for Disaster Risk Reduction

UN-ESCAP: United Nations Economic and Social Commission for Asia and the Pacific